1 $\lambda_E^{\leq}$ (Seal)

1.1 Syntax

$$
\begin{aligned}
A, B, C, R, S, T, \Gamma ::= {} & \mathsf{Int} \mid \varepsilon \mid A \to B \mid A \,\&\, B \mid \{\ell : A\} \mid \alpha_n \\
\mid {} & A \vee B \mid X_k \mid \mu T \mid \beta_i \mid \forall.\, A \\
\Delta \;\in {} & \mathbb{N} \rightharpoonup \mathsf{Typ} \qquad \emptyset \;=\; \lambda n.\ \mathsf{none} \\
e ::= {} & ? \mid e.k \mid i \mid \varepsilon \mid \lambda(A{\to}B).\, e \mid e_1 \triangleright e_2 \mid \langle e_1, \lambda(A{\to}B).\, e_2 \rangle \mid e_1\, e_2 \\
\mid {} & e_1 \,\#\, e_2 \mid \{\ell = e\} \mid e.\ell \mid (e : A) \mid \langle n \rangle\, e \mid \mathsf{seal}\ n\ R\ S\ e \mid \mathsf{unseal}\ n\ R\ S\ e \\
\mid {} & \mathsf{inl}^B\, e \mid \mathsf{inr}^A\, e \mid \mathsf{case}\ e\ \mathsf{of}\ e_1 \mid e_2 \mid \mathsf{fold}^T e \mid \mathsf{unfold}\ e \\
\mid {} & \mu\lambda(A{\to}B).\, e \mid \langle e_1, \mu\lambda(A{\to}B).\, e_2 \rangle^{B_x} \mid \Lambda_A.\, e \mid \langle e_1, \Lambda_A.\, e_2 \rangle \mid e\,[\alpha_n] \\
v ::= {} & i \mid \varepsilon \mid \langle v, \lambda(A{\to}B).\, e \rangle \mid \{\ell = v\} \mid v_1 \,\#\, v_2 \mid \langle n \rangle\, v \\
\mid {} & \mathsf{inl}^B\, v \mid \mathsf{inr}^A\, v \mid \mathsf{fold}^T v \\
\mid {} & \langle v, \mu\lambda(A{\to}B).\, e \rangle^{B_x} \mid \langle v, \Lambda_A.\, e \rangle
\end{aligned}
$$

X_k is a μ-bound type variable and β_i a $\forall$-bound *brand* variable (both de Bruijn, disjoint namespaces from each other and from concrete brands α_n); $\Lambda_A.\, e$ and its closure carry the $\forall$-body annotation A (casting at a $\forall$-type is an identity that the value must pin).

Three substitutions, all shift-free because the substitutend is closed in the relevant namespace: brand substitution $S[n{:=}R]$ (replace α_n by R), μ-unfolding $T[X_d{:=}S]^\mu$ (replace X_d by S; $d{+}1$ under μ), and $\forall\beta$-instantiation $A[\beta_d{:=}\alpha_n]$ (replace β_d by the *concrete* α_n; $d{+}1$ under $\forall$; μ binds X, not β — no bump). Each is homomorphic on all other formers; the interesting leaves:

$$
\begin{aligned}
&\alpha_m[n{:=}R] = \begin{cases} R & m = n \\ \alpha_m & m \neq n \end{cases} \qquad X_k[n{:=}R] = X_k \qquad \beta_i[n{:=}R] = \beta_i \\
&(\mu T)[n{:=}R] = \mu\, T[n{:=}R] \qquad (\forall.\, A)[n{:=}R] = \forall.\, A[n{:=}R] \\
&X_k[X_d{:=}S]^\mu = \begin{cases} S & k = d \\ X_k & k \neq d \end{cases} \qquad \beta_i[X_d{:=}S]^\mu = \beta_i \qquad \alpha_m[X_d{:=}S]^\mu = \alpha_m \\
&\beta_i[\beta_d{:=}\alpha_n] = \begin{cases} \alpha_n & i = d \\ \beta_i & i \neq d \end{cases} \qquad X_k[\beta_d{:=}\alpha_n] = X_k \\
&(\mu T)[\beta_d{:=}\alpha_n] = \mu\, T[\beta_d{:=}\alpha_n] \qquad (\forall.\, A)[\beta_d{:=}\alpha_n] = \forall.\, A[\beta_{d+1}{:=}\alpha_n]
\end{aligned}
$$

$e[\beta_d{:=}\alpha_n]$ pushes the instantiation into every type annotation of e ($d{+}1$ under Λ and its closure's annotation/body; a captured environment sits outside the binder and substitutes at d).

Occurrence predicates: $\mathsf{bv}(T)$ (some β_i occurs in T, index-blind); $\forall \in T$ (some $\forall$ occurs in T); $\mathsf{BClosed}\ c\ T$ (every β_i in T is bound below the cutoff: $i < c$ plus the enclosing binders; $c{+}1$ under $\forall$). Size measures: tsize on types and esize on terms (annotations excluded); both are preserved by $\forall\beta$-instantiation, which only trades a β_i leaf for a α_n leaf — the well-foundedness of the logical relation and of the fundamental lemma's induction rests on this.

$$
\frac{}{A\&B \ni_0 B}\ \textsc{zero} \qquad \frac{A \ni_k C}{A\&B \ni_{k+1} C}\ \textsc{succ} \qquad \frac{}{\ell \in \{\ell : A\}}\ \textsc{rcd} \qquad \frac{\ell \in A}{\ell \in A\&B}\ \textsc{andl} \qquad \frac{\ell \in B}{\ell \in A\&B}\ \textsc{andr}
$$

$$
\frac{}{\ell : A \in \{\ell : A\}}\ \textsc{zero} \qquad \frac{\ell : C \in A \qquad \neg\, \ell \in B}{\ell : C \in A\&B}\ \textsc{landl} \qquad \frac{\ell : C \in B \qquad \neg\, \ell \in A}{\ell : C \in A\&B}\ \textsc{landr}
$$

$$
\frac{}{\mathsf{Int}\ \mathsf{ord}}\ \textsc{oint} \qquad \frac{}{A \to B\ \mathsf{ord}}\ \textsc{oarr} \qquad \frac{}{\{\ell : A\}\ \mathsf{ord}}\ \textsc{orcd} \qquad \frac{}{\alpha_n\ \mathsf{ord}}\ \textsc{obrand} \qquad \frac{}{A \vee B\ \mathsf{ord}}\ \textsc{oor}
$$

$$
\frac{}{X_k\ \mathsf{ord}}\ \textsc{ovar} \qquad \frac{}{\mu T\ \mathsf{ord}}\ \textsc{omu} \qquad \frac{}{\beta_i\ \mathsf{ord}}\ \textsc{obvar} \qquad \frac{}{\forall.\, A\ \mathsf{ord}}\ \textsc{oallb} \qquad \frac{}{\rceil \varepsilon \lceil}\ \textsc{tltop} \qquad \frac{\rceil A \lceil \qquad \rceil B \lceil}{\rceil A\&B \lceil}\ \textsc{tland}
$$

$$
\frac{\rceil B \lceil}{\rceil A \to B \lceil}\ \textsc{tlarr} \qquad \frac{\rceil B \lceil}{\rceil \{\ell : B\} \lceil}\ \textsc{tlrcd}
$$

$$\mathsf{Int}^\uparrow = \varepsilon \quad \varepsilon^\uparrow = \varepsilon \quad (A \to B)^\uparrow = \langle \varepsilon, \lambda(A{\to}B).\, B^\uparrow \rangle \quad (A\&B)^\uparrow = A^\uparrow \,\#\, B^\uparrow \quad \{\ell : A\}^\uparrow = \{\ell = A^\uparrow\}$$
$$\alpha_n{}^\uparrow = (A \vee B)^\uparrow = X_k{}^\uparrow = (\mu T)^\uparrow = \beta_i{}^\uparrow = (\forall.\, A)^\uparrow = \varepsilon$$

1.2 Typing

Subtyping $A <: B$

$$\frac{}{\mathsf{Int} <: \mathsf{Int}}\ \textsc{sint} \qquad \frac{}{A <: \varepsilon}\ \textsc{stop} \qquad \frac{B_1 <: A_1 \qquad A_2 <: B_2}{A_1 \to A_2 <: B_1 \to B_2}\ \textsc{sarr} \qquad \frac{A_1 <: A_3}{A_1 \& A_2 <: A_3}\ \textsc{sandl}$$

$$\frac{A_2 <: A_3}{A_1 \& A_2 <: A_3}\ \textsc{sandr} \qquad \frac{A_1 <: A_2 \qquad A_1 <: A_3}{A_1 <: A_2 \& A_3}\ \textsc{sand} \qquad \frac{A <: B}{\{\ell : A\} <: \{\ell : B\}}\ \textsc{srcd} \qquad \frac{}{\alpha_n <: \alpha_n}\ \textsc{sbrand}$$

$$\frac{A_1 <: B_1 \qquad A_2 <: B_2}{A_1 \vee A_2 <: B_1 \vee B_2}\ \textsc{sor} \qquad \frac{}{X_k <: X_k}\ \textsc{svar} \qquad \frac{}{\mu T <: \mu T}\ \textsc{smu} \qquad \frac{}{\beta_i <: \beta_i}\ \textsc{sbvar} \qquad \frac{}{\forall.\, A <: \forall.\, A}\ \textsc{sallb}$$

Unions are component-wise only (no injection $A <: A \vee B$: casting would have to invent a tag); μ, X, β and $\forall$ are opaque, reflexive-only. *No* rule $\alpha_n <: R$: that would be translucency and kills representation independence.

Common ordinary supertypes and disjointness $A \sqcap B$, $A * B \;:=\; \neg(A \sqcap B)$

$$\frac{}{\mathsf{Int} \sqcap \mathsf{Int}}\ \textsc{cint} \qquad \frac{A \sqcap C}{A\&B \sqcap C}\ \textsc{candl} \qquad \frac{B \sqcap C}{A\&B \sqcap C}\ \textsc{candr} \qquad \frac{A \sqcap B}{A \sqcap B\&C}\ \textsc{crandl} \qquad \frac{A \sqcap C}{A \sqcap B\&C}\ \textsc{crandr}$$

$$\frac{B \sqcap D}{A \to B \sqcap C \to D}\ \textsc{carr} \qquad \frac{A \sqcap B}{\{\ell : A\} \sqcap \{\ell : B\}}\ \textsc{crcd} \qquad \frac{}{\alpha_n \sqcap \alpha_n}\ \textsc{cbrand} \qquad \frac{}{A \vee B \sqcap A' \vee B'}\ \textsc{coror}$$

$$\frac{}{X_k \sqcap X_k}\ \textsc{cvar} \qquad \frac{}{\mu T \sqcap \mu T}\ \textsc{cmu} \qquad \frac{}{\beta_i \sqcap \beta_j}\ \textsc{cbvar} \qquad \frac{}{\beta_i \sqcap \alpha_n}\ \textsc{cbvarbrand} \qquad \frac{}{\alpha_n \sqcap \beta_i}\ \textsc{cbrandbvar}$$

$$\frac{}{\forall.\, A \sqcap \forall.\, A}\ \textsc{callb} \qquad \frac{\mathsf{bv}(A) \vee \mathsf{bv}(A')}{\forall.\, A \sqcap \forall.\, A'}\ \textsc{callbbv} \qquad \frac{\mathsf{bv}(T) \vee \mathsf{bv}(T')}{\mu T \sqcap \mu T'}\ \textsc{cmubv}$$

The brand-variable rules are deliberately *conservative*: a β_i may be instantiated to any brand, so it costs every β_j and every α_n (unconditionally, in both directions), and two syntactically distinct $\forall$- or μ-bodies containing brand variables can become equal after instantiation (CALLBBV/CMUBV). This is exactly what makes disjointness stable under instantiation: $A * B \Rightarrow A[\beta_d{:=}\alpha_n] * B[\beta_d{:=}\alpha_n]$, with no freshness side condition anywhere (disj_bsubst). The price: nothing typed at a brand variable (or a β-containing $\mu/\forall$) can be *merged* with anything brand- or β-typed — the same wrap-it-in-records discipline unions already impose via COROR.

Casting and consistency $v \hookrightarrow_A v', \quad v_1 \approx v_2 \;:=\; \forall A\, w_1\, w_2.\; v_1 \hookrightarrow_A w_1 \Rightarrow v_2 \hookrightarrow_A w_2 \Rightarrow w_1 = w_2$

$$\frac{}{i \hookrightarrow_{\mathsf{Int}} i}\ \textsc{cint} \qquad \frac{}{v \hookrightarrow_{\varepsilon} \varepsilon}\ \textsc{ctop} \qquad \frac{\neg\rceil D\lceil \qquad C <: A \qquad B <: D}{\langle v, \lambda(A{\to}B).e\rangle \hookrightarrow_{C\to D} \langle v, \lambda(A{\to}D).e\rangle}\ \textsc{carrow}$$

$$\frac{\rceil D\lceil \qquad C <: A \qquad B <: D}{\langle v, \lambda(A{\to}B).e\rangle \hookrightarrow_{C\to D} \langle \varepsilon, \lambda(C{\to}D).D^{\uparrow}\rangle}\ \textsc{carrowtl} \qquad \frac{A\ \mathsf{ord} \qquad v_1 \hookrightarrow_A v_1'}{v_1 \# v_2 \hookrightarrow_A v_1'}\ \textsc{cmrgl}$$

$$\frac{A\ \mathsf{ord} \qquad v_2 \hookrightarrow_A v_2'}{v_1 \# v_2 \hookrightarrow_A v_2'}\ \textsc{cmrgr} \qquad \frac{v \hookrightarrow_A v_1 \qquad v \hookrightarrow_B v_2}{v \hookrightarrow_{A\&B} v_1 \# v_2}\ \textsc{cand} \qquad \frac{v \hookrightarrow_A v'}{\{\ell = v\} \hookrightarrow_{\{\ell:A\}} \{\ell = v'\}}\ \textsc{crcd}$$

$$\frac{}{\langle n\rangle\, v \hookrightarrow_{\alpha_n} \langle n\rangle\, v}\ \textsc{cwrap} \qquad \frac{v \hookrightarrow_{A'} v'}{\mathsf{inl}^{B} v \hookrightarrow_{A'\vee B'} \mathsf{inl}^{B'} v'}\ \textsc{cinl} \qquad \frac{v \hookrightarrow_{B'} v'}{\mathsf{inr}^{A} v \hookrightarrow_{A'\vee B'} \mathsf{inr}^{A'} v'}\ \textsc{cinr}$$

$$\frac{}{\mathsf{fold}^{T} v \hookrightarrow_{\mu T} \mathsf{fold}^{T} v}\ \textsc{cfold} \qquad \frac{\neg\rceil D\lceil \qquad C <: A \qquad B_x <: D}{\langle v, \mu\lambda(A{\to}B).\, e\rangle^{B_x} \hookrightarrow_{C\to D} \langle v, \mu\lambda(A{\to}B).\, e\rangle^{D}}\ \textsc{cfarrow}$$

$$\frac{\rceil D\lceil \qquad C <: A \qquad B_x <: D}{\langle v, \mu\lambda(A{\to}B).\, e\rangle^{B_x} \hookrightarrow_{C\to D} \langle \varepsilon, \lambda(C{\to}D).D^{\uparrow}\rangle}\ \textsc{cfarrowtl} \qquad \frac{}{\langle \rho, \Lambda_A.\, e\rangle \hookrightarrow_{\forall.\, A} \langle \rho, \Lambda_A.\, e\rangle}\ \textsc{cbclos}$$

Casting follows the union tag; at μ- and $\forall$-types it is the identity on folds of the same body resp. brand closures with the same annotation (the CWRAP pattern), and *undefined* at X_k and β_i (no cast rule — a term whose redex casts at a brand variable is stuck until instantiated). A fixpoint closure carries two codomains: casts rewrite only the external B_x; the internal B is pinned because beta reinstalls the self-reference at it.

Reserved-label freedom and signature well-formedness $\mathsf{NoRes}\ T, \quad \mathsf{WfSig}\ n\ R\ S$

$$\frac{}{\mathsf{NoRes\ Int}} \qquad \frac{}{\mathsf{NoRes}\ \varepsilon} \qquad \frac{}{\mathsf{NoRes}\ \alpha_n} \qquad \frac{\mathsf{NoRes}\ A \qquad \mathsf{NoRes}\ B}{\mathsf{NoRes}\ (A \to B)} \qquad \frac{\ell \neq \#\mathtt{f} \qquad \mathsf{NoRes}\ A}{\mathsf{NoRes}\ \{\ell : A\}}$$

$$\frac{\mathsf{NoRes}\ A \qquad \mathsf{NoRes}\ B}{\mathsf{NoRes}\ (A\&B)} \qquad \frac{\mathsf{NoRes}\ A \qquad \mathsf{NoRes}\ B}{\mathsf{NoRes}\ (A \vee B)} \qquad \frac{}{\mathsf{NoRes}\ X_k} \qquad \frac{\mathsf{NoRes}\ T}{\mathsf{NoRes}\ (\mu T)} \qquad \frac{}{\mathsf{NoRes}\ \beta_i} \qquad \frac{\mathsf{NoRes}\ A}{\mathsf{NoRes}\ (\forall.\, A)}$$

$$\frac{}{\mathsf{WfSig}\ n\ R\ \mathsf{Int}} \qquad \frac{}{\mathsf{WfSig}\ n\ R\ \varepsilon} \qquad \frac{}{\mathsf{WfSig}\ n\ R\ \alpha_m} \qquad \frac{\mathsf{WfSig}\ n\ R\ A \qquad \mathsf{WfSig}\ n\ R\ B}{\mathsf{WfSig}\ n\ R\ (A \to B)}$$

$$\frac{\ell \neq \#\mathtt{f} \qquad \mathsf{WfSig}\ n\ R\ A}{\mathsf{WfSig}\ n\ R\ \{\ell : A\}} \qquad \frac{\mathsf{WfSig}\ n\ R\ A \qquad \mathsf{WfSig}\ n\ R\ B \qquad A * B \qquad A[n{:=}R] * B[n{:=}R]}{\mathsf{WfSig}\ n\ R\ (A\&B)}$$

$$\frac{\mathsf{WfSig}\ n\ R\ A \qquad \mathsf{WfSig}\ n\ R\ B}{\mathsf{WfSig}\ n\ R\ (A \vee B)} \qquad \frac{\alpha_n \notin \mu T \qquad \mathsf{NoRes}\ (\mu T) \qquad \neg\mathsf{bv}(\mu T)}{\mathsf{WfSig}\ n\ R\ (\mu T)}$$

$$\frac{\alpha_n \notin \forall.\, A \qquad \mathsf{NoRes}\ (\forall.\, A) \qquad \neg\mathsf{bv}(\forall.\, A)}{\mathsf{WfSig}\ n\ R\ (\forall.\, A)}$$

WfSig has *no* X_k and no β_i case: signatures are brand-variable-free throughout (wfsig_no_hasbvar), so sealing annotations are invariant under $\forall\beta$-instantiation. μ- and $\forall$-components of a signature are opaque under sealing (the coercion is the identity on them), hence must not mention the sealed brand α_n.

Typing $\Delta; \Gamma \vdash e : A$

$$\frac{}{\Delta;\Gamma \vdash ? : \Gamma}\ \textsc{tquery} \qquad \frac{}{\Delta;\Gamma \vdash i : \mathsf{Int}}\ \textsc{tint} \qquad \frac{}{\Delta;\Gamma \vdash \varepsilon : \varepsilon}\ \textsc{tunit} \qquad \frac{\Delta;\Gamma \vdash e_1 : A \to B \qquad \Delta;\Gamma \vdash e_2 : A}{\Delta;\Gamma \vdash e_1\, e_2 : B}\ \textsc{tapp}$$

$$\frac{\Delta;\Gamma \vdash e_1 : \Gamma_1 \qquad \Delta;\Gamma_1 \vdash e_2 : A}{\Delta;\Gamma \vdash e_1 \triangleright e_2 : A}\ \textsc{tbox} \qquad \frac{\Delta;\Gamma \vdash e : B \qquad B \ni_k A}{\Delta;\Gamma \vdash e.k : A}\ \textsc{tproj} \qquad \frac{\Delta;\Gamma \vdash e : A}{\Delta;\Gamma \vdash \{\ell = e\} : \{\ell : A\}}\ \textsc{trcd}$$

$$\frac{\Delta;\Gamma \vdash e : B \qquad \ell : A \in B}{\Delta;\Gamma \vdash e.\ell : A}\ \textsc{trproj} \qquad \frac{\Delta;\Gamma \vdash e_1 : A \qquad \Delta;\Gamma\&A \vdash e_2 : B \qquad A * \Gamma \qquad A * B}{\Delta;\Gamma \vdash e_1 \# e_2 : A\&B}\ \textsc{tmrg}$$

$$\frac{v_1\ \mathsf{value} \qquad v_2\ \mathsf{value} \qquad \Delta;\varepsilon \vdash v_1 : A \qquad \Delta;\varepsilon \vdash v_2 : B \qquad A * B \qquad v_1 \approx v_2}{\Delta;\Gamma \vdash v_1 \# v_2 : A\&B}\ \textsc{tmergev}$$

$$\frac{\begin{array}{c} v_1\ \mathsf{value} \qquad v_2\ \mathsf{value} \qquad \Delta;\varepsilon \vdash v_1 : A \qquad \Delta;\varepsilon \vdash v_2 : B \\ v\ \mathsf{value} \qquad \Delta;\varepsilon \vdash v : C \qquad v \hookrightarrow_A v_1 \qquad v \hookrightarrow_B v_2 \qquad v_1 \approx v_2 \end{array}}{\Delta;\Gamma \vdash v_1 \# v_2 : A\&B}\ \textsc{tmergevc} \qquad \frac{\Gamma * A \qquad \Delta;\Gamma\&A \vdash e : B}{\Delta;\Gamma \vdash \lambda(A{\to}B).e : A \to B}\ \textsc{tlam}$$

$$\frac{v\ \mathsf{value} \qquad \Delta;\varepsilon \vdash v : \Gamma_1 \qquad \Gamma_1 * A \qquad \Delta;\Gamma_1\&A \vdash e : B' \qquad B' <: B \qquad C <: A}{\Delta;\Gamma \vdash \langle v, \lambda(A{\to}B).e\rangle : C \to B}\ \textsc{tclos}$$

$$\frac{\Delta;\Gamma \vdash e : B \qquad B <: A}{\Delta;\Gamma \vdash (e : A) : A}\ \textsc{tanno} \qquad \frac{\Delta\, n = R \qquad \neg\mathsf{bv}(R) \qquad v\ \mathsf{value} \qquad \Delta;\varepsilon \vdash v : R}{\Delta;\Gamma \vdash \langle n\rangle\, v : \alpha_n}\ \textsc{twrap}$$

$$\frac{\Delta\, n = R \qquad \mathsf{NoRes}\ R \qquad \neg\mathsf{bv}(R) \qquad \mathsf{WfSig}\ n\ R\ S \qquad \Delta;\Gamma \vdash e : S[n{:=}R]}{\Delta;\Gamma \vdash \mathsf{seal}\ n\ R\ S\ e : S}\ \textsc{tseal}$$

$$\frac{\Delta\, n = R \qquad \mathsf{NoRes}\ R \qquad \neg\mathsf{bv}(R) \qquad \mathsf{WfSig}\ n\ R\ S \qquad \Delta;\Gamma \vdash e : S}{\Delta;\Gamma \vdash \mathsf{unseal}\ n\ R\ S\ e : S[n{:=}R]}\ \textsc{tunseal} \qquad \frac{\Delta;\Gamma \vdash e : A}{\Delta;\Gamma \vdash \mathsf{inl}^B e : A \vee B}\ \textsc{tinl}$$

$$\frac{\Delta;\Gamma \vdash e : B}{\Delta;\Gamma \vdash \mathsf{inr}^A e : A \vee B}\ \textsc{tinr}$$

$$\frac{\Delta;\Gamma \vdash e : A \vee B \qquad \Gamma * A \qquad \Gamma * B \qquad \Delta;\Gamma\&A \vdash e_1 : C \qquad \Delta;\Gamma\&B \vdash e_2 : C}{\Delta;\Gamma \vdash \mathsf{case}\ e\ \mathsf{of}\ e_1 \mid e_2 : C}\ \textsc{tcase}$$

$$\frac{\neg\forall \in T \qquad \Delta;\Gamma \vdash e : T[X_0{:=}\mu T]^\mu}{\Delta;\Gamma \vdash \mathsf{fold}^T e : \mu T}\ \textsc{tfold} \qquad \frac{\neg\forall \in T \qquad \Delta;\Gamma \vdash e : \mu T \qquad A = T[X_0{:=}\mu T]^\mu}{\Delta;\Gamma \vdash \mathsf{unfold}\ e : A}\ \textsc{tunfold}$$

$$\frac{\Gamma * (A{\to}B) \qquad (\Gamma\&(A{\to}B)) * A \qquad \Delta;(\Gamma\&(A{\to}B))\&A \vdash e : B}{\Delta;\Gamma \vdash \mu\lambda(A{\to}B).e : A \to B}\ \textsc{tflam}$$

$$\frac{\begin{array}{c} v\ \mathsf{value} \qquad \Delta;\varepsilon \vdash v : \Gamma_1 \qquad \Gamma_1 * (A{\to}B) \qquad (\Gamma_1\&(A{\to}B)) * A \\ \Delta;(\Gamma_1\&(A{\to}B))\&A \vdash e : B' \qquad B' <: B \qquad B <: B_x \qquad C <: A \end{array}}{\Delta;\Gamma \vdash \langle v, \mu\lambda(A{\to}B).\, e\rangle^{B_x} : C \to B_x}\ \textsc{tfclos} \qquad \frac{\neg\mathsf{bv}(\Gamma) \qquad \Delta;\Gamma \vdash e : A}{\Delta;\Gamma \vdash \Lambda_A.\, e : \forall.\, A}\ \textsc{tblam}$$

$$\frac{\rho\ \mathsf{value} \qquad \Delta;\varepsilon \vdash \rho : \Gamma_1 \qquad \neg\mathsf{bv}(\Gamma_1) \qquad \Delta;\Gamma_1 \vdash e : A}{\Delta;\Gamma \vdash \langle \rho, \Lambda_A.\, e\rangle : \forall.\, A}\ \textsc{tbclos}$$

$$\frac{\Delta;\Gamma \vdash e : \forall.\, A \qquad \mathsf{BClosed}\ 1\ A \qquad T = A[\beta_0{:=}\alpha_n]}{\Delta;\Gamma \vdash e\,[\alpha_n] : T}\ \textsc{tbapp}$$

TMERGEV is the module-composition merge (disjoint components); TMERGEVC is the cast-duplicated merge

produced by casting one value at $A \,\&\, B$, recording the common origin v — that record is what keeps the rule stable under $\forall\beta$-instantiation (consistency of the substituted components is rebuilt from casts of the substituted origin rather than assumed). The three $\forall$-fences: $\neg\forall{\in}T$ on TFOLD/TUNFOLD (μ-unfolding does not shift brand variables); $\mathsf{BClosed}\ 1\ A$ on TBAPP (the shift-free beta is index-coherent only for single-binder bodies); $\neg\mathsf{bv}(\cdot)$ on captured contexts and representations (they must be instantiation-invariant).

$$\Delta \sqsubseteq \Delta' \;:=\; \forall n\, R.\ \Delta\, n = R \Rightarrow \Delta'\, n = R$$

1.3 Semantics

Value lookups $\quad v \ni_k v', \quad \ell : v' \in v$

$$\frac{}{v_1 \,\#\, v_2 \ni_0 v_2}\ \text{LVZERO} \qquad \frac{v_1 \ni_k v_3}{v_1 \,\#\, v_2 \ni_{k+1} v_3}\ \text{LVSUCC} \qquad \frac{}{\ell : e \in \{\ell = e\}}\ \text{RVLZERO} \qquad \frac{\ell : e \in e_1}{\ell : e \in e_1 \,\#\, e_2}\ \text{VLANDL}$$

$$\frac{\ell : e \in e_2}{\ell : e \in e_1 \,\#\, e_2}\ \text{VLANDR}$$

Sealing coercions $\quad v \hookrightarrow_S^{\mathsf{seal}\, n\, R} w, \quad v \hookrightarrow_S^{\mathsf{unseal}\, n\, R} w \qquad (\mathsf{pxe}(c) := \varepsilon \,\#\, \{\#\mathtt{f} = c\}, \quad \mathsf{pxf} := (?.1).\#\mathtt{f})$

$$\frac{}{v \hookrightarrow_{\alpha_n}^{\mathsf{seal}\, n\, R} \langle n \rangle\, v}\ \text{BRAND-EQ} \qquad \frac{m \neq n}{v \hookrightarrow_{\alpha_m}^{\mathsf{seal}\, n\, R} v}\ \text{BRAND-NE} \qquad \frac{}{i \hookrightarrow_{\mathsf{Int}}^{\mathsf{seal}\, n\, R} i}\ \text{INT} \qquad \frac{}{v \hookrightarrow_{\varepsilon}^{\mathsf{seal}\, n\, R} \varepsilon}\ \text{TOP}$$

$$\frac{v_1 \hookrightarrow_A^{\mathsf{seal}\, n\, R} w_1 \qquad v_2 \hookrightarrow_B^{\mathsf{seal}\, n\, R} w_2}{v_1 \,\#\, v_2 \hookrightarrow_{A\&B}^{\mathsf{seal}\, n\, R} w_1 \,\#\, w_2}\ \text{AND} \qquad \frac{v \hookrightarrow_A^{\mathsf{seal}\, n\, R} w}{\{\ell = v\} \hookrightarrow_{\{\ell:A\}}^{\mathsf{seal}\, n\, R} \{\ell = w\}}\ \text{RCD}$$

$$\frac{}{c \hookrightarrow_{A\to B}^{\mathsf{seal}\, n\, R} \langle \mathsf{pxe}(c), \lambda(A{\to}B).\ \mathsf{seal}\ n\ R\ B\ (\mathsf{pxf}\ (\mathsf{unseal}\ n\ R\ A\ ?.0))\rangle}\ \text{ARR} \qquad \frac{v \hookrightarrow_A^{\mathsf{seal}\, n\, R} w}{\mathsf{inl}^{B[n:=R]} v \hookrightarrow_{A\vee B}^{\mathsf{seal}\, n\, R} \mathsf{inl}^{B} w}\ \text{INL}$$

$$\frac{v \hookrightarrow_B^{\mathsf{seal}\, n\, R} w}{\mathsf{inr}^{A[n:=R]} v \hookrightarrow_{A\vee B}^{\mathsf{seal}\, n\, R} \mathsf{inr}^{A} w}\ \text{INR} \qquad \frac{}{v \hookrightarrow_{X_k}^{\mathsf{seal}\, n\, R} v}\ \text{VAR} \qquad \frac{}{v \hookrightarrow_{\mu T}^{\mathsf{seal}\, n\, R} v}\ \text{MU} \qquad \frac{}{v \hookrightarrow_{\beta_i}^{\mathsf{seal}\, n\, R} v}\ \text{BVAR}$$

$$\frac{}{v \hookrightarrow_{\forall.\, A}^{\mathsf{seal}\, n\, R} v}\ \text{ALLB}$$

$$\frac{}{\langle n \rangle\, v \hookrightarrow_{\alpha_n}^{\mathsf{unseal}\, n\, R} v}\ \text{BRAND-EQ} \qquad \frac{m \neq n}{v \hookrightarrow_{\alpha_m}^{\mathsf{unseal}\, n\, R} v}\ \text{BRAND-NE} \qquad \frac{}{i \hookrightarrow_{\mathsf{Int}}^{\mathsf{unseal}\, n\, R} i}\ \text{INT} \qquad \frac{}{v \hookrightarrow_{\varepsilon}^{\mathsf{unseal}\, n\, R} \varepsilon}\ \text{TOP}$$

$$\frac{v_1 \hookrightarrow_A^{\mathsf{unseal}\, n\, R} w_1 \qquad v_2 \hookrightarrow_B^{\mathsf{unseal}\, n\, R} w_2}{v_1 \,\#\, v_2 \hookrightarrow_{A\&B}^{\mathsf{unseal}\, n\, R} w_1 \,\#\, w_2}\ \text{AND} \qquad \frac{v \hookrightarrow_A^{\mathsf{unseal}\, n\, R} w}{\{\ell = v\} \hookrightarrow_{\{\ell:A\}}^{\mathsf{unseal}\, n\, R} \{\ell = w\}}\ \text{RCD}$$

$$\frac{}{c \hookrightarrow_{A\to B}^{\mathsf{unseal}\, n\, R} \langle \mathsf{pxe}(c), \lambda(A[n:=R]{\to}B[n:=R]).\ \mathsf{unseal}\ n\ R\ B\ (\mathsf{pxf}\ (\mathsf{seal}\ n\ R\ A\ ?.0))\rangle}\ \text{ARR}$$

$$\frac{v \hookrightarrow_A^{\mathsf{unseal}\, n\, R} w}{\mathsf{inl}^{B} v \hookrightarrow_{A\vee B}^{\mathsf{unseal}\, n\, R} \mathsf{inl}^{B[n:=R]} w}\ \text{INL} \qquad \frac{v \hookrightarrow_B^{\mathsf{unseal}\, n\, R} w}{\mathsf{inr}^{A} v \hookrightarrow_{A\vee B}^{\mathsf{unseal}\, n\, R} \mathsf{inr}^{A[n:=R]} w}\ \text{INR} \qquad \frac{}{v \hookrightarrow_{X_k}^{\mathsf{unseal}\, n\, R} v}\ \text{VAR}$$

$$\frac{}{v \hookrightarrow_{\mu T}^{\mathsf{unseal}\, n\, R} v}\ \text{MU} \qquad \frac{}{v \hookrightarrow_{\beta_i}^{\mathsf{unseal}\, n\, R} v}\ \text{BVAR} \qquad \frac{}{v \hookrightarrow_{\forall.\, A}^{\mathsf{unseal}\, n\, R} v}\ \text{ALLB}$$

The coercions are the identity at $X/\mu/\beta/\forall$ (opaque under sealing; type-correct because WfSig forbids α_n and brand variables there).

Small-step $v \vdash e \hookrightarrow e'$ **(all rules carry** v value**; value premises shown)**

$$\frac{}{v \vdash ? \hookrightarrow v}\ \textsc{squery} \qquad \frac{v \vdash e_1 \hookrightarrow e_1'}{v \vdash e_1\, e_2 \hookrightarrow e_1'\, e_2}\ \textsc{sappl} \qquad \frac{v_1\ \mathsf{value} \qquad v \vdash e_2 \hookrightarrow e_2'}{v \vdash v_1\, e_2 \hookrightarrow v_1\, e_2'}\ \textsc{sappr}$$

$$\frac{v_1\ \mathsf{value} \qquad v_2\ \mathsf{value} \qquad v_2 \hookrightarrow_A v_2'}{v \vdash \langle v_1, \lambda(A{\to}B).e\rangle\, v_2 \hookrightarrow (v_1 \# v_2') \triangleright (e : B)}\ \textsc{sbeta} \qquad \frac{}{v \vdash \lambda(A{\to}B).e \hookrightarrow \langle v, \lambda(A{\to}B).e\rangle}\ \textsc{sclos}$$

$$\frac{v \vdash e_1 \hookrightarrow e_1'}{v \vdash e_1 \triangleright e_2 \hookrightarrow e_1' \triangleright e_2}\ \textsc{sboxl} \qquad \frac{v_1\ \mathsf{value} \qquad v_1 \vdash e_2 \hookrightarrow e_2'}{v \vdash v_1 \triangleright e_2 \hookrightarrow v_1 \triangleright e_2'}\ \textsc{sboxr} \qquad \frac{v_1\ \mathsf{value} \qquad v_2\ \mathsf{value}}{v \vdash v_1 \triangleright v_2 \hookrightarrow v_2}\ \textsc{sboxv}$$

$$\frac{v \vdash e_1 \hookrightarrow e_1'}{v \vdash e_1 \# e_2 \hookrightarrow e_1' \# e_2}\ \textsc{smrgl} \qquad \frac{v_1\ \mathsf{value} \qquad v \# v_1 \vdash e_2 \hookrightarrow e_2'}{v \vdash v_1 \# e_2 \hookrightarrow v_1 \# e_2'}\ \textsc{smrgr} \qquad \frac{v \vdash e \hookrightarrow e'}{v \vdash e.k \hookrightarrow e'.k}\ \textsc{sproj}$$

$$\frac{v_1\ \mathsf{value} \qquad v_1 \ni_k v'}{v \vdash v_1.k \hookrightarrow v'}\ \textsc{sprojv} \qquad \frac{v \vdash e \hookrightarrow e'}{v \vdash \{\ell = e\} \hookrightarrow \{\ell = e'\}}\ \textsc{slrec} \qquad \frac{v \vdash e \hookrightarrow e'}{v \vdash e.\ell \hookrightarrow e'.\ell}\ \textsc{srproj}$$

$$\frac{v_1\ \mathsf{value} \qquad \ell : v' \in v_1}{v \vdash v_1.\ell \hookrightarrow v'}\ \textsc{srprojv} \qquad \frac{v \vdash e \hookrightarrow e'}{v \vdash (e : A) \hookrightarrow (e' : A)}\ \textsc{sanno} \qquad \frac{v_1\ \mathsf{value} \qquad v_1 \hookrightarrow_A v'}{v \vdash (v_1 : A) \hookrightarrow v'}\ \textsc{sannov}$$

$$\frac{v \vdash e \hookrightarrow e'}{v \vdash \langle n\rangle\, e \hookrightarrow \langle n\rangle\, e'}\ \textsc{swrap} \qquad \frac{v \vdash e \hookrightarrow e'}{v \vdash \mathsf{seal}\ n\ R\ S\ e \hookrightarrow \mathsf{seal}\ n\ R\ S\ e'}\ \textsc{sseal} \qquad \frac{v_1\ \mathsf{value} \qquad v_1 \hookrightarrow_S^{\mathsf{seal}\, n\, R} w}{v \vdash \mathsf{seal}\ n\ R\ S\ v_1 \hookrightarrow w}\ \textsc{sealv}$$

$$\frac{v \vdash e \hookrightarrow e'}{v \vdash \mathsf{unseal}\ n\ R\ S\ e \hookrightarrow \mathsf{unseal}\ n\ R\ S\ e'}\ \textsc{sunseal} \qquad \frac{v_1\ \mathsf{value} \qquad v_1 \hookrightarrow_S^{\mathsf{unseal}\, n\, R} w}{v \vdash \mathsf{unseal}\ n\ R\ S\ v_1 \hookrightarrow w}\ \textsc{sunsealv}$$

$$\frac{v \vdash e \hookrightarrow e'}{v \vdash \mathsf{inl}^B e \hookrightarrow \mathsf{inl}^B e'}\ \textsc{sinl} \qquad \frac{v \vdash e \hookrightarrow e'}{v \vdash \mathsf{inr}^A e \hookrightarrow \mathsf{inr}^A e'}\ \textsc{sinr} \qquad \frac{v \vdash e \hookrightarrow e'}{v \vdash \mathsf{case}\ e\ \mathsf{of}\ e_1 \mid e_2 \hookrightarrow \mathsf{case}\ e'\ \mathsf{of}\ e_1 \mid e_2}\ \textsc{scase}$$

$$\frac{v_1\ \mathsf{value}}{v \vdash \mathsf{case}\ (\mathsf{inl}^B v_1)\ \mathsf{of}\ e_1 \mid e_2 \hookrightarrow (v \# v_1) \triangleright e_1}\ \textsc{scasel} \qquad \frac{v_1\ \mathsf{value}}{v \vdash \mathsf{case}\ (\mathsf{inr}^A v_1)\ \mathsf{of}\ e_1 \mid e_2 \hookrightarrow (v \# v_1) \triangleright e_2}\ \textsc{scaser}$$

$$\frac{v \vdash e \hookrightarrow e'}{v \vdash \mathsf{fold}^T e \hookrightarrow \mathsf{fold}^T e'}\ \textsc{sfold} \qquad \frac{v \vdash e \hookrightarrow e'}{v \vdash \mathsf{unfold}\ e \hookrightarrow \mathsf{unfold}\ e'}\ \textsc{sunfold} \qquad \frac{v_1\ \mathsf{value}}{v \vdash \mathsf{unfold}\ (\mathsf{fold}^T v_1) \hookrightarrow v_1}\ \textsc{sunfoldv}$$

$$\frac{}{v \vdash \mu\lambda(A{\to}B).e \hookrightarrow \langle v, \mu\lambda(A{\to}B).\, e\rangle^B}\ \textsc{sflam}$$

$$\frac{v_1\ \mathsf{value} \qquad v_2\ \mathsf{value} \qquad v_2 \hookrightarrow_A v_2'}{v \vdash \langle v_1, \mu\lambda(A{\to}B).\, e\rangle^{B_x}\, v_2 \hookrightarrow \big((v_1 \# \langle v_1, \mu\lambda(A{\to}B).\, e\rangle^B) \# v_2'\big) \triangleright (e : B_x)}\ \textsc{sfbeta}$$

$$\frac{}{v \vdash \Lambda_A.\, e \hookrightarrow \langle v, \Lambda_A.\, e\rangle}\ \textsc{sblam} \qquad \frac{v \vdash e \hookrightarrow e'}{v \vdash e\,[\alpha_n] \hookrightarrow e'\,[\alpha_n]}\ \textsc{sbappl} \qquad \frac{\rho\ \mathsf{value}}{v \vdash \langle \rho, \Lambda_A.\, e\rangle\,[\alpha_n] \hookrightarrow \rho \triangleright e[\beta_0{:=}\alpha_n]}\ \textsc{sbappbeta}$$

Fixpoint beta reinstalls the self-reference with the external codomain reset to the internal B and reseals the result at the client-facing B_x. $\forall$-beta is pure annotation rewriting: no argument cast, no reseal — the instantiated body runs under the captured environment.

$$\frac{}{v \vdash e \hookrightarrow^* e}\ \textsc{refl} \qquad \frac{v \vdash e_1 \hookrightarrow e_2 \qquad v \vdash e_2 \hookrightarrow^* e_3}{v \vdash e_1 \hookrightarrow^* e_3}\ \textsc{step}$$

1.4 Meta-theory

Theorem 1.1 (gdeterminism)**.**

$$\Delta;\Gamma \vdash e : A \ \wedge\ \Delta;\varepsilon \vdash v : \Gamma \ \wedge\ v \vdash e \hookrightarrow e_1 \ \wedge\ v \vdash e \hookrightarrow e_2 \implies e_1 = e_2$$

Theorem 1.2 (gprogress)**.**

$$\Delta;\Gamma \vdash e : A \ \wedge\ v\ \mathsf{value} \ \wedge\ \Delta;\varepsilon \vdash v : \Gamma \implies e\ \mathsf{value} \ \vee\ \exists e'.\ v \vdash e \hookrightarrow e'$$

Theorem 1.3 (gpreservation)**.**

$$v \vdash e \hookrightarrow e' \ \wedge\ \Delta;\Gamma \vdash e : A \ \wedge\ \Delta;\varepsilon \vdash v : \Gamma \implies \Delta;\Gamma \vdash e' : A$$

Theorem 1.4 (hastype_weaken_store)**.**

$$\Delta \sqsubseteq \Delta' \ \wedge\ \Delta;\Gamma \vdash e : A \implies \Delta';\Gamma \vdash e : A$$

Definition 1.5 (viewL, viewR, WrapRel, RawRel, LRg, LR)**.** $o \in \{\mathsf{none}\} \cup \mathbb{N} \times \mathsf{Typ} \times \mathsf{Typ}$*;* $\eta : \mathbb{N} \to \mathsf{Exp} \to \mathsf{Exp} \to \mathsf{Prop}$*;* *stores* Δ_1, Δ_2 *ambient.*

$$\begin{array}{ll} \mathsf{viewL}\ \mathsf{none}\, T = T & \mathsf{viewL}\ (n, R_1, R_2)\, T = T[n{:=}R_1] \\ \mathsf{viewR}\ \mathsf{none}\, T = T & \mathsf{viewR}\ (n, R_1, R_2)\, T = T[n{:=}R_2] \end{array}$$

$$\begin{aligned} \mathsf{WrapRel}\ m\ (v_1, v_2) &= \Delta_1;\varepsilon \vdash v_1 : \alpha_m \wedge \Delta_2;\varepsilon \vdash v_2 : \alpha_m \wedge \exists w_1\, w_2.\ v_1 = \langle m \rangle\, w_1 \wedge v_2 = \langle m \rangle\, w_2 \wedge \eta\, m\, w_1\, w_2 \\ \mathsf{RawRel}\ n\ R_1\ R_2\ (v_1, v_2) &= v_1\ \mathsf{value} \wedge v_2\ \mathsf{value} \wedge \Delta_1;\varepsilon \vdash v_1 : R_1 \wedge \Delta_2;\varepsilon \vdash v_2 : R_2 \wedge \eta\, n\, v_1\, v_2 \end{aligned}$$

$$\begin{aligned} \mathcal{V}[\![\mathsf{Int}]\!]^o(v_1, v_2) &= \exists i.\ v_1 = i \wedge v_2 = i \\ \mathcal{V}[\![\varepsilon]\!]^o(v_1, v_2) &= v_1 = \varepsilon \wedge v_2 = \varepsilon \\ \mathcal{V}[\![\alpha_m]\!]^o(v_1, v_2) &= \begin{cases} \mathsf{RawRel}\ n\ R_1\ R_2\ (v_1, v_2) & o = (n, R_1, R_2),\ m = n \\ \mathsf{WrapRel}\ m\ (v_1, v_2) & \textit{otherwise} \end{cases} \\ \mathcal{V}[\![A \& B]\!]^o(v_1, v_2) &= \Delta_1;\varepsilon \vdash v_1 : \mathsf{viewL}\, o\, (A \& B) \wedge \Delta_2;\varepsilon \vdash v_2 : \mathsf{viewR}\, o\, (A \& B) \wedge \\ &\quad \exists a_1\, b_1\, a_2\, b_2.\ v_1 = a_1 \,\#\, b_1 \wedge v_2 = a_2 \,\#\, b_2 \wedge \mathcal{V}[\![A]\!]^o(a_1, a_2) \wedge \mathcal{V}[\![B]\!]^o(b_1, b_2) \\ \mathcal{V}[\![\{\ell : A\}]\!]^o(v_1, v_2) &= \exists w_1\, w_2.\ v_1 = \{\ell = w_1\} \wedge v_2 = \{\ell = w_2\} \wedge \mathcal{V}[\![A]\!]^o(w_1, w_2) \\ \mathcal{V}[\![A \to B]\!]^o(v_1, v_2) &= v_1\ \mathsf{value} \wedge v_2\ \mathsf{value} \wedge \Delta_1;\varepsilon \vdash v_1 : \mathsf{viewL}\, o\, (A{\to}B) \wedge \Delta_2;\varepsilon \vdash v_2 : \mathsf{viewR}\, o\, (A{\to}B) \wedge \\ &\quad \forall u_1\, u_2.\ \mathcal{V}[\![A]\!]^o(u_1, u_2) \Rightarrow \forall \rho_1\, \rho_2.\ \rho_1\ \mathsf{value} \Rightarrow \rho_2\ \mathsf{value} \Rightarrow \\ &\qquad \exists w_1\, w_2.\ \rho_1 \vdash v_1\, u_1 \hookrightarrow^* w_1 \wedge \rho_2 \vdash v_2\, u_2 \hookrightarrow^* w_2 \wedge \mathcal{V}[\![B]\!]^o(w_1, w_2) \end{aligned}$$

$$\begin{aligned} \mathcal{V}[\![A \vee B]\!]^o(v_1, v_2) &= \big(\exists w_1\, w_2.\ v_1 = \mathsf{inl}^{\mathsf{viewL}\, o\, B} w_1 \wedge v_2 = \mathsf{inl}^{\mathsf{viewR}\, o\, B} w_2 \wedge \mathcal{V}[\![A]\!]^o(w_1, w_2)\big) \vee \\ &\quad \big(\exists w_1\, w_2.\ v_1 = \mathsf{inr}^{\mathsf{viewL}\, o\, A} w_1 \wedge v_2 = \mathsf{inr}^{\mathsf{viewR}\, o\, A} w_2 \wedge \mathcal{V}[\![B]\!]^o(w_1, w_2)\big) \\ \mathcal{V}[\![X_k]\!]^o(v_1, v_2) &= \bot \qquad\qquad \mathcal{V}[\![\beta_i]\!]^o(v_1, v_2) \ =\ \bot \\ \mathcal{V}[\![\mu T]\!]^o(v_1, v_2) &= v_1\ \mathsf{value} \wedge v_2\ \mathsf{value} \wedge \Delta_1;\varepsilon \vdash v_1 : \mathsf{viewL}\, o\, (\mu T) \wedge \Delta_2;\varepsilon \vdash v_2 : \mathsf{viewR}\, o\, (\mu T) \wedge v_1 = v_2 \\ \mathcal{V}[\![\forall.\, A]\!]^o(v_1, v_2) &= v_1\ \mathsf{value} \wedge v_2\ \mathsf{value} \wedge \Delta_1;\varepsilon \vdash v_1 : \mathsf{viewL}\, o\, (\forall.\, A) \wedge \Delta_2;\varepsilon \vdash v_2 : \mathsf{viewR}\, o\, (\forall.\, A) \wedge \\ &\quad \forall m.\ \forall \rho_1\, \rho_2.\ \rho_1\ \mathsf{value} \Rightarrow \rho_2\ \mathsf{value} \Rightarrow \\ &\qquad \exists w_1\, w_2.\ \rho_1 \vdash v_1\, [\alpha_m] \hookrightarrow^* w_1 \wedge \rho_2 \vdash v_2\, [\alpha_m] \hookrightarrow^* w_2 \wedge \mathcal{V}[\![A[\beta_0{:=}\alpha_m]]\!]^o(w_1, w_2) \end{aligned}$$

$$\mathcal{V}[\![T]\!] \ :=\ \mathcal{V}[\![T]\!]^{\mathsf{none}}$$

The $\forall$*-clause is* instantiation closure*: related generic values yield, at every concrete brand* m*, related runs of the instantiation at the substituted body type.* η *is never extended and no freshness is demanded — at brand-*m *positions the relation is whatever* η *already says about* m*, which is exactly what representation independence needs (the client instantiates at the provider's own allocated brand, where* η *relates the two representations). Typings stay at the* syntactic *type, so degenerate* β*-annotated values relate vacuously through the* $\bot$ *clause at* β_i*. The recursion descends on* tsize *(*$A[\beta_0{:=}\alpha_m]$ *is not a subterm but is size-preserving), so the definition is by well-founded recursion.* μ *stays opaque (a finer clause needs step indexing); free theorems over* arbitrary *relations at* β *(relational parametricity proper) are future work.*

Definition 1.6 (Finitary)**.** Finitary *e: the least predicate closed under every term former* except *the fixpoints* $\mu\lambda(A{\to}B).e$ *and their closures. Generic terms* $\Lambda/\langle\cdot, \Lambda\rangle/\cdot\,[\alpha_n]$ are *included:* $\forall$*-beta rewrites only annotations and never duplicates subterms, so the fragment stays normalizing. The relational layer below is stated for this fragment; the safety layer (determinism, progress, preservation) covers the full language.*

Definition 1.7 (OpenAdm, OpenBrand, SemTyp)**.**

$$\mathsf{OpenAdm}\ \eta\ o = \forall n\, R_1\, R_2.\ o = (n, R_1, R_2) \Rightarrow \neg\rceil R_1 \lceil \wedge \neg\rceil R_2 \lceil \wedge \\ (\forall w_1\, w_2.\ \eta\, n\, w_1\, w_2 \Rightarrow \forall w_1'\, w_2'.\ w_1 \hookrightarrow_{R_1} w_1' \Rightarrow w_2 \hookrightarrow_{R_2} w_2' \Rightarrow \eta\, n\, w_1'\, w_2')$$

$$\mathsf{OpenBrand}\ \Delta_1\ \Delta_2\ \eta\ n\ R_1\ R_2\ =\ \begin{cases} \Delta_1\, n = R_1\ \wedge\ \Delta_2\, n = R_2\ \wedge\ \mathsf{NoRes}\ R_1\ \wedge\ \mathsf{NoRes}\ R_2\ \wedge\ \neg\rceil R_1 \lceil \wedge \neg\rceil R_2 \lceil \wedge \\ \forall w_1\, w_2.\ \eta\, n\, w_1\, w_2 \Rightarrow \forall w_1'\, w_2'.\ w_1 \hookrightarrow_{R_1} w_1' \Rightarrow w_2 \hookrightarrow_{R_2} w_2' \Rightarrow \eta\, n\, w_1'\, w_2' \end{cases}$$

$$\mathsf{SemTyp}\ \Delta_1\ \Delta_2\ \eta\ \Gamma\ e_1\ e_2\ A\ =\ \forall \rho_1\, \rho_2.\ \mathcal{V}[\![\Gamma]\!](\rho_1, \rho_2) \Rightarrow \exists w_1\, w_2.\ \rho_1 \vdash e_1 \hookrightarrow^* w_1 \wedge \rho_2 \vdash e_2 \hookrightarrow^* w_2 \wedge \mathcal{V}[\![A]\!](w_1, w_2)$$

Theorem 1.8 (fundamental)**.**

$$\emptyset; \Gamma \vdash e : A\ \wedge\ \mathsf{Finitary}\ e\ \Longrightarrow\ \mathsf{SemTyp}\ \Delta_1\ \Delta_2\ \eta\ \Gamma\ e\ e\ A$$

Proved by strong induction on size e*: the* Λ*-cases recurse at the* instantiated *body — the unsubstituted body cannot run (casts at* β *annotations are stuck) — whose typing comes from the instantiation lemma* $\Delta; \Gamma \vdash e : A \Rightarrow \Delta; \Gamma[\beta_d{:=}\alpha_n] \vdash e[\beta_d{:=}\alpha_n] : A[\beta_d{:=}\alpha_n]$ *(hastype_bsubst), not from a subderivation.*

Theorem 1.9 (normalization)**.**

$$\emptyset; \varepsilon \vdash e : A\ \wedge\ \mathsf{Finitary}\ e\ \Longrightarrow\ \exists v.\ v\ \mathsf{value}\ \wedge\ \varepsilon \vdash e \hookrightarrow^* v$$

Theorem 1.10 (sealing)**.**

$$\mathcal{V}[\![A]\!](p_1', p_2')\ \wedge\ \emptyset; A \vdash e : \mathsf{Int}\ \wedge\ \mathsf{Finitary}\ e\ \Longrightarrow\ \exists i.\ p_1' \vdash e \hookrightarrow^* i\ \wedge\ p_2' \vdash e \hookrightarrow^* i$$

Theorem 1.11 (coe_lr)**.**

$$\mathsf{OpenBrand}\ \Delta_1\ \Delta_2\ \eta\ n\ R_1\ R_2\ \wedge\ \mathsf{WfSig}\ n\ R_1\ S\ \wedge\ \mathsf{WfSig}\ n\ R_2\ S\ \Longrightarrow$$
$$\big(\forall v_1\, v_2\, w_1\, w_2.\ \mathcal{V}[\![S]\!]^{(n,R_1,R_2)}(v_1, v_2) \Rightarrow v_1 \hookrightarrow_S^{\mathsf{seal}\, n\, R_1} w_1 \Rightarrow v_2 \hookrightarrow_S^{\mathsf{seal}\, n\, R_2} w_2 \Rightarrow \mathcal{V}[\![S]\!](w_1, w_2)\big) \wedge$$
$$\big(\forall v_1\, v_2\, w_1\, w_2.\ \mathcal{V}[\![S]\!](v_1, v_2) \Rightarrow v_1 \hookrightarrow_S^{\mathsf{unseal}\, n\, R_1} w_1 \Rightarrow v_2 \hookrightarrow_S^{\mathsf{unseal}\, n\, R_2} w_2 \Rightarrow \mathcal{V}[\![S]\!]^{(n,R_1,R_2)}(w_1, w_2)\big)$$

Theorem 1.12 (representation_independence)**.**

$$\mathsf{OpenBrand}\ \Delta_1\ \Delta_2\ \eta\ n\ R_1\ R_2\ \wedge\ \mathsf{WfSig}\ n\ R_1\ S\ \wedge\ \mathsf{WfSig}\ n\ R_2\ S\ \wedge \\ \mathcal{V}[\![S]\!]^{(n,R_1,R_2)}(p_1, p_2)\ \wedge\ \emptyset; S \vdash e : \mathsf{Int}\ \wedge\ \mathsf{Finitary}\ e \\ \Longrightarrow\ \exists i.\ \varepsilon \vdash (\mathsf{seal}\ n\ R_1\ S\ p_1) \triangleright e \hookrightarrow^* i\ \wedge\ \varepsilon \vdash (\mathsf{seal}\ n\ R_2\ S\ p_2) \triangleright e \hookrightarrow^* i$$

Theorem 1.13 (generic_client_representation_independence)**.** *One* generic *client, typed once at* $\forall.\,(S_\beta \to \mathsf{Int})$ *with the signature abstract in the brand it will be instantiated at, observes the same integer against two sealed providers of that signature over different representations. Writing* $S^* := S_\beta[\beta_0{:=}\alpha_n]$*:*

$$\mathsf{OpenBrand}\ \Delta_1\ \Delta_2\ \eta\ n\ R_1\ R_2\ \wedge\ \mathsf{WfSig}\ n\ R_1\ S^*\ \wedge\ \mathsf{WfSig}\ n\ R_2\ S^*\ \wedge \\ \mathcal{V}[\![S^*]\!]^{(n,R_1,R_2)}(p_1, p_2)\ \wedge\ \emptyset; \varepsilon \vdash c_t : \forall.\,(S_\beta \to \mathsf{Int})\ \wedge\ \mathsf{Finitary}\ c_t \\ \Longrightarrow\ \exists i.\ \varepsilon \vdash (c_t\,[\alpha_n])\,(\mathsf{seal}\ n\ R_1\ S^*\ p_1) \hookrightarrow^* i\ \wedge\ \varepsilon \vdash (c_t\,[\alpha_n])\,(\mathsf{seal}\ n\ R_2\ S^*\ p_2) \hookrightarrow^* i$$

The client instantiates at the provider's own *allocated brand* n*: the* $\forall$*-clause needs no freshness, and* η *at* n *is exactly the relation between the two representations, so* $\cdot\,[\alpha_n]$ *lands the client in the concrete-signature world where representation_independence already lives.*

2 λ_{SCE} (SCE)

2.1 Syntax

$$\begin{aligned} T, A, B, R, S, \Gamma &::= \mathsf{Int} \mid \varepsilon \mid T \to T \mid T \,\&\, T \mid T \vee T \mid \{\ell : T\} \mid \mathsf{sig}\ \sigma \mid X_k \mid \mu T \mid \alpha_n \\ \sigma &::= \langle T \rangle \mid T \Rightarrow \sigma \\ s &::= \bullet \mid \circ \qquad \Delta \in \mathbb{N} \rightharpoonup \mathsf{Typ} \qquad \emptyset = \lambda n.\,\mathsf{none} \qquad \Delta \sqsubseteq \Delta' := \forall n\, R.\ \Delta\, n = R \Rightarrow \Delta'\, n = R \\ e &::= ? \mid e.k \mid i \mid \varepsilon \mid \lambda A.\, e \mid e_1 \triangleright e_2 \mid \langle e_1, \lambda A.\, e_2 \rangle \mid e_1\, e_2 \mid e_1 \,\#\, e_2 \mid \{\ell = e\} \mid e.\ell \\ &\mid \mathsf{struct}^s\, e \mid \mathsf{functor}^s(A).\, e \mid \langle e_1, \mathsf{functor}(A).\, e_2 \rangle \mid e_1 \ltimes e_2 \mid e_1 \bullet e_2 \mid e_1 \ltimes^* e_2 \\ &\mid e_1 \,,, e_2 \mid \mathsf{let}^A\, e_1\ \mathsf{in}\ e_2 \mid \mathsf{open}\ e_1\ \mathsf{in}\ e_2 \\ &\mid \mathsf{inl}^B\, e \mid \mathsf{inr}^A\, e \mid \mathsf{case}\ e\ \mathsf{of}\ e_1 \mid e_2 \mid \mu\lambda(A{\to}B).\, e \mid \langle e_1, \mu\lambda(A{\to}B).\, e_2 \rangle \mid \mathsf{fold}^T e \mid \mathsf{unfold}\ e \\ &\mid \langle n \rangle\, e \mid \mathsf{seal}\ n\ R\ S\ e \mid \mathsf{unseal}\ n\ R\ S\ e \\ v &::= i \mid \varepsilon \mid \langle v, \lambda A.e \rangle \mid \langle v, \mathsf{functor}(A).e \rangle \mid v_1 \,\#\, v_2 \mid \{\ell = v\} \\ &\mid \mathsf{inl}^B v \mid \mathsf{inr}^A v \mid \langle v, \mu\lambda(A{\to}B).e \rangle \mid \mathsf{fold}^T v \mid \langle n \rangle\, v \end{aligned}$$

$$X_k[d{:=}S]^\mu = \begin{cases} S & k = d \\ X_k & k \neq d \end{cases} \qquad (\mu T)[d{:=}S]^\mu = \mu\,(T[d{+}1{:=}S]^\mu) \qquad \alpha_n[d{:=}S]^\mu = \alpha_n$$

$$\alpha_m[n{:=}R] = \begin{cases} R & m = n \\ \alpha_m & m \neq n \end{cases} \qquad X_k[n{:=}R] = X_k \qquad (\mu T)[n{:=}R] = \mu\,(T[n{:=}R])$$

(both homomorphic otherwise, incl. σ)

$$\frac{}{A\&B \ni_0 B}\ \text{ZERO} \qquad \frac{A \ni_k C}{A\&B \ni_{k+1} C}\ \text{SUCC} \qquad \frac{}{\ell \in \{\ell : T\}}\ \text{RCD} \qquad \frac{\ell \in A}{\ell \in A\&B}\ \text{ANDL} \qquad \frac{\ell \in B}{\ell \in A\&B}\ \text{ANDR}$$

$$\frac{\ell \in T}{\ell \in \mathsf{sig}\,\langle T\rangle}\ \text{SIG} \qquad \frac{}{\ell : T \in \{\ell : T\}}\ \text{ZERO} \qquad \frac{\ell : T \in A \qquad \neg\,\ell \in B}{\ell : T \in A\&B}\ \text{ANDL} \qquad \frac{\ell : T \in B \qquad \neg\,\ell \in A}{\ell : T \in A\&B}\ \text{ANDR}$$

$$\frac{\ell : A \in T}{\ell : A \in \mathsf{sig}\,\langle T\rangle}\ \text{SIG}$$

2.2 Elaboration (SCE $\to \lambda_E^{\leq}$)

Type translation $\quad [\![\cdot]\!] : \mathsf{SCE.Typ} \to \mathsf{Seal.Typ}, \quad [\![\Delta]\!] := \lambda n.\ [\![\Delta\, n]\!]$

$$[\![\mathsf{Int}]\!] = \mathsf{Int} \quad [\![\varepsilon]\!] = \varepsilon \quad [\![A \to B]\!] = [\![A]\!] \to [\![B]\!] \quad [\![A\&B]\!] = [\![A]\!]\,\&\,[\![B]\!] \quad [\![A \vee B]\!] = \varepsilon \quad [\![\{\ell : A\}]\!] = \{\ell : [\![A]\!]\}$$
$$[\![\mathsf{sig}\,\langle T\rangle]\!] = [\![T]\!] \quad [\![\mathsf{sig}\,(T \Rightarrow \sigma)]\!] = [\![T]\!] \to [\![\mathsf{sig}\,\sigma]\!] \quad [\![X_k]\!] = \varepsilon \quad [\![\mu T]\!] = \varepsilon \quad [\![\alpha_n]\!] = \alpha_n$$

Import packages $\quad \mathsf{wire}(\Gamma, c_1, D), \quad \mathsf{WireOk}\ \Gamma\ \Gamma_1\ D$

$$\mathsf{wire}(\Gamma, c_1, \{\ell : A\}) = \{\ell = c_1.\ell\} \qquad \mathsf{wire}(\Gamma, c_1, _) = \varepsilon$$
$$\mathsf{wire}(\Gamma, c_1, D\&\{\ell : A\}) = \mathsf{wire}(\Gamma, c_1, D)\ \#\ \big((? : \Gamma) \triangleright \{\ell = c_1.\ell\}\big)$$

$$\frac{\ell : A \in \Gamma_1}{\mathsf{WireOk}\ \Gamma\ \Gamma_1\ \{\ell : A\}}\ \text{ONE} \qquad \frac{\mathsf{WireOk}\ \Gamma\ \Gamma_1\ D \qquad \ell : A \in \Gamma_1 \qquad [\![D]\!] * \Gamma \qquad [\![D]\!] * \{\ell : [\![A]\!]\}}{\mathsf{WireOk}\ \Gamma\ \Gamma_1\ (D\&\{\ell : A\})}\ \text{MORE}$$

Elaboration $\Delta; \Gamma \vdash e : A \rightsquigarrow \overline{e}$

$$\frac{}{\Delta; \Gamma \vdash ? : \Gamma \rightsquigarrow ?}\ \textsc{equery} \qquad \frac{}{\Delta; \Gamma \vdash i : \mathsf{Int} \rightsquigarrow i}\ \textsc{elit} \qquad \frac{}{\Delta; \Gamma \vdash \varepsilon : \varepsilon \rightsquigarrow \varepsilon}\ \textsc{eunit}$$

$$\frac{\Delta; \Gamma \vdash e_1 : A \to B \rightsquigarrow \overline{e}_1 \qquad \Delta; \Gamma \vdash e_2 : A \rightsquigarrow \overline{e}_2}{\Delta; \Gamma \vdash e_1\, e_2 : B \rightsquigarrow \overline{e}_1\, \overline{e}_2}\ \textsc{eapp} \qquad \frac{\Delta; \Gamma \vdash e : A \rightsquigarrow \overline{e} \qquad A \ni_k B}{\Delta; \Gamma \vdash e.k : B \rightsquigarrow \overline{e}.k}\ \textsc{eproj}$$

$$\frac{\Delta; \Gamma \vdash e_1 : \Gamma' \rightsquigarrow \overline{e}_1 \qquad \Delta; \Gamma' \vdash e_2 : A \rightsquigarrow \overline{e}_2}{\Delta; \Gamma \vdash e_1 \triangleright e_2 : A \rightsquigarrow \overline{e}_1 \triangleright \overline{e}_2}\ \textsc{ebox}$$

$$\frac{\Delta; \Gamma \vdash e_1 : A \rightsquigarrow \overline{e}_1 \qquad \Delta; \Gamma \& A \vdash e_2 : B \rightsquigarrow \overline{e}_2 \qquad [\![A]\!] * [\![\Gamma]\!] \qquad [\![A]\!] * [\![B]\!]}{\Delta; \Gamma \vdash e_1 \# e_2 : A \& B \rightsquigarrow \overline{e}_1 \# \overline{e}_2}\ \textsc{edmrg}$$

$$\frac{v_1\ \mathsf{value} \qquad v_2\ \mathsf{value} \qquad \Delta; \varepsilon \vdash v_1 : A \rightsquigarrow \overline{w}_1 \qquad \Delta; \varepsilon \vdash v_2 : B \rightsquigarrow \overline{w}_2 \qquad [\![A]\!] * [\![B]\!]}{\Delta; \Gamma \vdash v_1 \# v_2 : A \& B \rightsquigarrow \overline{w}_1 \# \overline{w}_2}\ \textsc{evmrg}$$

$$\frac{\Delta; \Gamma \vdash e_1 : A \rightsquigarrow \overline{e}_1 \qquad \Delta; \Gamma \vdash e_2 : B \rightsquigarrow \overline{e}_2 \qquad [\![A]\!] * [\![\Gamma]\!] \qquad [\![A]\!] * [\![B]\!]}{\Delta; \Gamma \vdash e_1 \,,, e_2 : A \& B \rightsquigarrow \overline{e}_1 \# \left((? : [\![\Gamma]\!]) \triangleright \overline{e}_2\right)}\ \textsc{enmrg}$$

$$\frac{\Delta; \Gamma \& A \vdash e : B \rightsquigarrow \overline{e} \qquad [\![\Gamma]\!] * [\![A]\!]}{\Delta; \Gamma \vdash \lambda A.e : A \to B \rightsquigarrow \lambda([\![A]\!] \to [\![B]\!]).\overline{e}}\ \textsc{elam} \qquad \frac{\Delta; \Gamma \vdash e : B \rightsquigarrow \overline{e} \qquad \ell : A \in B}{\Delta; \Gamma \vdash e.\ell : A \rightsquigarrow \overline{e}.\ell}\ \textsc{erproj}$$

$$\frac{v_1\ \mathsf{value} \qquad \Delta; \varepsilon \vdash v_1 : \Gamma' \rightsquigarrow \overline{v}_1 \qquad \Delta; \Gamma' \& A \vdash e_2 : B \rightsquigarrow \overline{e}_2 \qquad [\![\Gamma']\!] * [\![A]\!]}{\Delta; \Gamma \vdash \langle v_1, \lambda A.e_2 \rangle : A \to B \rightsquigarrow \langle \overline{v}_1, \lambda([\![A]\!] \to [\![B]\!]).\overline{e}_2 \rangle}\ \textsc{eclos}$$

$$\frac{\Delta; \Gamma \vdash e : A \rightsquigarrow \overline{e}}{\Delta; \Gamma \vdash \{\ell = e\} : \{\ell : A\} \rightsquigarrow \{\ell = \overline{e}\}}\ \textsc{elrec} \qquad \frac{\Delta; \Gamma \vdash e_1 : A \rightsquigarrow \overline{e}_1 \qquad \Delta; \Gamma \& A \vdash e_2 : B \rightsquigarrow \overline{e}_2 \qquad [\![\Gamma]\!] * [\![A]\!]}{\Delta; \Gamma \vdash \mathsf{let}^A e_1\ \mathsf{in}\ e_2 : B \rightsquigarrow \left(\lambda([\![A]\!] \to [\![B]\!]).\overline{e}_2\right) \overline{e}_1}\ \textsc{eletb}$$

$$\frac{\Delta; \Gamma \vdash e_1 : \{\ell : A\} \rightsquigarrow \overline{e}_1 \qquad \Delta; \Gamma \& A \vdash e_2 : B \rightsquigarrow \overline{e}_2 \qquad [\![\Gamma]\!] * [\![A]\!]}{\Delta; \Gamma \vdash \mathsf{open}\ e_1\ \mathsf{in}\ e_2 : B \rightsquigarrow \left(\lambda([\![A]\!] \to [\![B]\!]).\overline{e}_2\right) (\overline{e}_1.\ell)}\ \textsc{eopenm}$$

$$\frac{\Delta; \Gamma \vdash e : A \rightsquigarrow \overline{e}}{\Delta; \Gamma \vdash \mathsf{inl}^B e : A \vee B \rightsquigarrow \mathsf{inl}^{[\![B]\!]}\, \overline{e}}\ \textsc{einl} \qquad \frac{\Delta; \Gamma \vdash e : B \rightsquigarrow \overline{e}}{\Delta; \Gamma \vdash \mathsf{inr}^A e : A \vee B \rightsquigarrow \mathsf{inr}^{[\![A]\!]}\, \overline{e}}\ \textsc{einr}$$

$$\frac{\Delta; \Gamma \vdash e : A \vee B \rightsquigarrow \overline{e} \qquad \Delta; \Gamma \& A \vdash e_1 : C \rightsquigarrow \overline{e}_1 \qquad \Delta; \Gamma \& B \vdash e_2 : C \rightsquigarrow \overline{e}_2 \qquad [\![\Gamma]\!] * [\![A]\!] \qquad [\![\Gamma]\!] * [\![B]\!]}{\Delta; \Gamma \vdash \mathsf{case}\ e\ \mathsf{of}\ e_1 \mid e_2 : C \rightsquigarrow \mathsf{case}\ \overline{e}\ \mathsf{of}\ \overline{e}_1 \mid \overline{e}_2}\ \textsc{ecase}$$

$$\frac{\Delta; (\Gamma \& (A \to B)) \& A \vdash e : B \rightsquigarrow \overline{e} \qquad [\![\Gamma]\!] * ([\![A]\!] \to [\![B]\!]) \qquad ([\![\Gamma]\!] \& ([\![A]\!] \to [\![B]\!])) * [\![A]\!]}{\Delta; \Gamma \vdash \mu\lambda(A \to B).e : A \to B \rightsquigarrow \mu\lambda([\![A]\!] \to [\![B]\!]).\overline{e}}\ \textsc{eflam}$$

$$\frac{\begin{array}{c} v_1\ \mathsf{value} \qquad \Delta; \varepsilon \vdash v_1 : \Gamma' \rightsquigarrow \overline{v}_1 \\ \Delta; (\Gamma' \& (A \to B)) \& A \vdash e_2 : B \rightsquigarrow \overline{e}_2 \qquad [\![\Gamma']\!] * ([\![A]\!] \to [\![B]\!]) \qquad ([\![\Gamma']\!] \& ([\![A]\!] \to [\![B]\!])) * [\![A]\!] \end{array}}{\Delta; \Gamma \vdash \langle v_1, \mu\lambda(A \to B).e_2 \rangle : A \to B \rightsquigarrow \langle \overline{v}_1, \mu\lambda([\![A]\!] \to [\![B]\!]).\, \overline{e}_2 \rangle^{[\![B]\!]}}\ \textsc{efclos}$$

$$\frac{\Delta; \Gamma \vdash e : T[X_0 := \mu T]^\mu \rightsquigarrow \overline{e}}{\Delta; \Gamma \vdash \mathsf{fold}^T e : \mu T \rightsquigarrow \mathsf{fold}^{[\![T]\!]}\, \overline{e}}\ \textsc{efold} \qquad \frac{\Delta; \Gamma \vdash e : \mu T \rightsquigarrow \overline{e} \qquad A = T[X_0 := \mu T]^\mu}{\Delta; \Gamma \vdash \mathsf{unfold}\ e : A \rightsquigarrow \mathsf{unfold}\ \overline{e}}\ \textsc{eunfold}$$

$$\frac{\Delta;\varepsilon \vdash e : B \rightsquigarrow \overline{e}}{\Delta;\Gamma \vdash \mathsf{struct}^{\bullet}\ e : B \rightsquigarrow \varepsilon \triangleright \overline{e}}\ \textsc{emstruct-}\bullet \qquad \frac{\Delta;\Gamma \vdash e : B \rightsquigarrow \overline{e}}{\Delta;\Gamma \vdash \mathsf{struct}^{\circ}\ e : B \rightsquigarrow ? \triangleright \overline{e}}\ \textsc{emstruct-}\circ$$

$$\frac{\Delta;\varepsilon \& A \vdash e : B \rightsquigarrow \overline{e}}{\Delta;\Gamma \vdash \mathsf{functor}^{\bullet}(A).e : \mathsf{sig}\,(A \Rightarrow \langle B\rangle) \rightsquigarrow \varepsilon \triangleright \lambda([\![A]\!]{\to}[\![B]\!]).\overline{e}}\ \textsc{emfunctor-}\bullet$$

$$\frac{\Delta;\Gamma \& A \vdash e : B \rightsquigarrow \overline{e} \qquad [\![\Gamma]\!] * [\![A]\!]}{\Delta;\Gamma \vdash \mathsf{functor}^{\circ}(A).e : \mathsf{sig}\,(A \Rightarrow \langle B\rangle) \rightsquigarrow \lambda([\![A]\!]{\to}[\![B]\!]).\overline{e}}\ \textsc{emfunctor-}\circ$$

$$\frac{v_1\ \mathsf{value} \qquad \Delta;\varepsilon \vdash v_1 : \Gamma' \rightsquigarrow \overline{v}_1 \qquad \Delta;\Gamma' \& A \vdash e_2 : B \rightsquigarrow \overline{e}_2 \qquad [\![\Gamma']\!] * [\![A]\!]}{\Delta;\Gamma \vdash \langle v_1, \mathsf{functor}(A).e_2\rangle : \mathsf{sig}\,(A \Rightarrow \langle B\rangle) \rightsquigarrow \langle \overline{v}_1, \lambda([\![A]\!]{\to}[\![B]\!]).\overline{e}_2\rangle}\ \textsc{emclos}$$

$$\frac{\Delta;\Gamma \vdash e_1 : \mathsf{sig}\,(A \Rightarrow \langle B\rangle) \rightsquigarrow \overline{e}_1 \qquad \Delta;\Gamma \vdash e_2 : A \rightsquigarrow \overline{e}_2}{\Delta;\Gamma \vdash e_1 \bullet e_2 : B \rightsquigarrow \overline{e}_1\,\overline{e}_2}\ \textsc{emapp}$$

$$\frac{\Delta;\Gamma \vdash e_1 : \Gamma_1 \rightsquigarrow \overline{e}_1 \qquad \Delta;\Gamma \vdash e_2 : \mathsf{sig}\,(\{\ell : A\} \Rightarrow \langle B\rangle) \rightsquigarrow \overline{e}_2 \qquad \ell : A \in \Gamma_1 \qquad [\![\Gamma_1]\!] * [\![\Gamma]\!] \qquad [\![\Gamma_1]\!] * [\![B]\!]}{\Delta;\Gamma \vdash e_1 \ltimes e_2 : \Gamma_1 \& B \rightsquigarrow \overline{e}_1 \,\#\, \Big((? : [\![\Gamma]\!]) \triangleright \big(\overline{e}_2\ \{\ell = \overline{e}_1.\ell\}\big)\Big)}\ \textsc{emlink}$$

$$\frac{\begin{array}{c}\Delta;\Gamma \vdash e_1 : \Gamma_1 \rightsquigarrow \overline{e}_1 \\ \Delta;\Gamma \vdash e_2 : \mathsf{sig}\,(D \Rightarrow \langle B\rangle) \rightsquigarrow \overline{e}_2 \qquad \mathsf{WireOk}\ [\![\Gamma]\!]\ \Gamma_1\ D \qquad [\![\Gamma_1]\!] * [\![\Gamma]\!] \qquad [\![\Gamma_1]\!] * [\![B]\!]\end{array}}{\Delta;\Gamma \vdash e_1 \ltimes^{*} e_2 : \Gamma_1 \& B \rightsquigarrow \overline{e}_1 \,\#\, \Big((? : [\![\Gamma]\!]) \triangleright \big(\overline{e}_2\ \mathsf{wire}([\![\Gamma]\!], \overline{e}_1, D)\big)\Big)}\ \textsc{emlinkn}$$

$$\frac{\Delta\, n = R \qquad v\ \mathsf{value} \qquad \Delta;\varepsilon \vdash v : R \rightsquigarrow \overline{w}}{\Delta;\Gamma \vdash \langle n\rangle\, v : \alpha_n \rightsquigarrow \langle n\rangle\, \overline{w}}\ \textsc{ewrap}$$

$$\frac{\Delta\, n = R \qquad \mathsf{NoRes}\ [\![R]\!] \qquad \mathsf{WfSig}\ n\ [\![R]\!]\ [\![S]\!] \qquad \Delta;\Gamma \vdash e : S[n{:=}R] \rightsquigarrow \overline{e}}{\Delta;\Gamma \vdash \mathsf{seal}\ n\ R\ S\ e : S \rightsquigarrow \mathsf{seal}\ n\ [\![R]\!]\ [\![S]\!]\ \overline{e}}\ \textsc{emseal}$$

$$\frac{\Delta\, n = R \qquad \mathsf{NoRes}\ [\![R]\!] \qquad \mathsf{WfSig}\ n\ [\![R]\!]\ [\![S]\!] \qquad \Delta;\Gamma \vdash e : S \rightsquigarrow \overline{e}}{\Delta;\Gamma \vdash \mathsf{unseal}\ n\ R\ S\ e : S[n{:=}R] \rightsquigarrow \mathsf{unseal}\ n\ [\![R]\!]\ [\![S]\!]\ \overline{e}}\ \textsc{emunseal}$$

2.3 Semantics

Value lookups, selection, packages $\quad v \ni_k v', \quad v.\ell \rightsquigarrow v', \quad v \gg_D p$

$$\frac{}{v_1 \,\#\, v_2 \ni_0 v_2}\ \textsc{dmrg-zero} \qquad \frac{v_1 \ni_k v'}{v_1 \,\#\, v_2 \ni_{k+1} v'}\ \textsc{dmrg-succ} \qquad \frac{}{v_1 \,,,\, v_2 \ni_0 v_2}\ \textsc{nmrg-zero}$$

$$\frac{v_1 \ni_k v'}{v_1 \,,,\, v_2 \ni_{k+1} v'}\ \textsc{nmrg-succ} \qquad \frac{}{\{\ell = v\}.\ell \rightsquigarrow v}\ \textsc{rcd} \qquad \frac{v_1.\ell \rightsquigarrow v'}{v_1 \,\#\, v_2.\ell \rightsquigarrow v'}\ \textsc{dmrg-left}$$

$$\frac{v_2.\ell \rightsquigarrow v'}{v_1 \,\#\, v_2.\ell \rightsquigarrow v'}\ \textsc{dmrg-right} \qquad \frac{v_1.\ell \rightsquigarrow v'}{v_1 \,,,\, v_2.\ell \rightsquigarrow v'}\ \textsc{nmrg-left} \qquad \frac{v_2.\ell \rightsquigarrow v'}{v_1 \,,,\, v_2.\ell \rightsquigarrow v'}\ \textsc{nmrg-right}$$

$$\frac{v.\ell \rightsquigarrow v_\ell}{v \gg_{\{\ell:A\}} \{\ell = v_\ell\}}\ \textsc{one} \qquad \frac{v \gg_D p \qquad v.\ell \rightsquigarrow v_\ell}{v \gg_{D\&\{\ell:A\}} p \,\#\, \{\ell = v_\ell\}}\ \textsc{more}$$

Sealing coercions $v \Downarrow_S^{\mathsf{seal}\, n\, R} w$, $v \Downarrow_S^{\mathsf{unseal}\, n\, R} w$ ($\mathsf{pxe}(c) := \varepsilon \# \{\#\mathtt{f} = c\}$, $\mathsf{pxf} := (?.1).\#\mathtt{f}$)

$$\frac{}{v \Downarrow_{\alpha_n}^{\mathsf{seal}\, n\, R} \langle n \rangle\, v}\ \text{BRAND-EQ} \qquad \frac{m \neq n}{v \Downarrow_{\alpha_m}^{\mathsf{seal}\, n\, R} v}\ \text{BRAND-NE} \qquad \frac{}{i \Downarrow_{\mathsf{Int}}^{\mathsf{seal}\, n\, R} i}\ \text{INT} \qquad \frac{}{v \Downarrow_{\varepsilon}^{\mathsf{seal}\, n\, R} \varepsilon}\ \text{TOP}$$

$$\frac{v_1 \Downarrow_A^{\mathsf{seal}\, n\, R} w_1 \qquad v_2 \Downarrow_B^{\mathsf{seal}\, n\, R} w_2}{v_1 \# v_2 \Downarrow_{A\&B}^{\mathsf{seal}\, n\, R} w_1 \# w_2}\ \text{AND} \qquad \frac{v \Downarrow_A^{\mathsf{seal}\, n\, R} w}{\{\ell = v\} \Downarrow_{\{\ell:A\}}^{\mathsf{seal}\, n\, R} \{\ell = w\}}\ \text{RCD}$$

$$\frac{}{c \Downarrow_{A\to B}^{\mathsf{seal}\, n\, R} \langle \mathsf{pxe}(c), \lambda A.\ \mathsf{seal}\ n\ R\ B\ (\mathsf{pxf}\ (\mathsf{unseal}\ n\ R\ A\ ?.0))\rangle}\ \text{ARR}$$

$$\frac{}{\langle n \rangle\, v \Downarrow_{\alpha_n}^{\mathsf{unseal}\, n\, R} v}\ \text{BRAND-EQ} \qquad \frac{m \neq n}{v \Downarrow_{\alpha_m}^{\mathsf{unseal}\, n\, R} v}\ \text{BRAND-NE} \qquad \frac{}{i \Downarrow_{\mathsf{Int}}^{\mathsf{unseal}\, n\, R} i}\ \text{INT} \qquad \frac{}{v \Downarrow_{\varepsilon}^{\mathsf{unseal}\, n\, R} \varepsilon}\ \text{TOP}$$

$$\frac{v_1 \Downarrow_A^{\mathsf{unseal}\, n\, R} w_1 \qquad v_2 \Downarrow_B^{\mathsf{unseal}\, n\, R} w_2}{v_1 \# v_2 \Downarrow_{A\&B}^{\mathsf{unseal}\, n\, R} w_1 \# w_2}\ \text{AND} \qquad \frac{v \Downarrow_A^{\mathsf{unseal}\, n\, R} w}{\{\ell = v\} \Downarrow_{\{\ell:A\}}^{\mathsf{unseal}\, n\, R} \{\ell = w\}}\ \text{RCD}$$

$$\frac{}{c \Downarrow_{A\to B}^{\mathsf{unseal}\, n\, R} \langle \mathsf{pxe}(c), \lambda A[n{:=}R].\ \mathsf{unseal}\ n\ R\ B\ (\mathsf{pxf}\ (\mathsf{seal}\ n\ R\ A\ ?.0))\rangle}\ \text{ARR}$$

Big-step evaluation $\rho \vdash e \Downarrow v$ **(all rules carry ρ value; value premises shown)**

$$\frac{}{\rho \vdash ? \Downarrow \rho}\ \textsc{query} \qquad \frac{}{\rho \vdash i \Downarrow i}\ \textsc{lit} \qquad \frac{}{\rho \vdash \varepsilon \Downarrow \varepsilon}\ \textsc{unit} \qquad \frac{v\ \mathsf{value}}{\rho \vdash \langle v, \lambda A.e\rangle \Downarrow \langle v, \lambda A.e\rangle}\ \textsc{clos-val}$$

$$\frac{v\ \mathsf{value}}{\rho \vdash \langle v, \mathsf{functor}(A).e\rangle \Downarrow \langle v, \mathsf{functor}(A).e\rangle}\ \textsc{mclos-val} \qquad \frac{\rho \vdash e \Downarrow v \qquad v \ni_k v'}{\rho \vdash e.k \Downarrow v'}\ \textsc{proj} \qquad \frac{}{\rho \vdash \lambda A.e \Downarrow \langle \rho, \lambda A.e\rangle}\ \textsc{lam}$$

$$\frac{\rho \vdash e_1 \Downarrow v_1 \qquad v_1 \vdash e_2 \Downarrow v}{\rho \vdash e_1 \triangleright e_2 \Downarrow v}\ \textsc{box} \qquad \frac{\rho \vdash e_1 \Downarrow \langle v_1, \lambda A.e\rangle \qquad \rho \vdash e_2 \Downarrow v_2 \qquad v_1 \# v_2 \vdash e \Downarrow v}{\rho \vdash e_1\, e_2 \Downarrow v}\ \textsc{app-clos}$$

$$\frac{\rho \vdash e_1 \Downarrow \langle v_1, \mu\lambda(A{\to}B).e\rangle \qquad \rho \vdash e_2 \Downarrow v_2 \qquad (v_1 \# \langle v_1, \mu\lambda(A{\to}B).e\rangle) \# v_2 \vdash e \Downarrow v}{\rho \vdash e_1\, e_2 \Downarrow v}\ \textsc{app-fclos}$$

$$\frac{\rho \vdash e_1 \Downarrow \langle v_1, \mathsf{functor}(A).e\rangle \qquad \rho \vdash e_2 \Downarrow v_2 \qquad v_1 \# v_2 \vdash e \Downarrow v}{\rho \vdash e_1 \bullet e_2 \Downarrow v}\ \textsc{app-mclos}$$

$$\frac{\rho \vdash e_1 \Downarrow v_1 \qquad \rho \# v_1 \vdash e_2 \Downarrow v_2}{\rho \vdash e_1 \# e_2 \Downarrow v_1 \# v_2}\ \textsc{dmrg} \qquad \frac{\rho \vdash e_1 \Downarrow v_1 \qquad \rho \vdash e_2 \Downarrow v_2}{\rho \vdash e_1 ,, e_2 \Downarrow v_1 \# v_2}\ \textsc{nmrg} \qquad \frac{\rho \vdash e \Downarrow v}{\rho \vdash \{\ell = e\} \Downarrow \{\ell = v\}}\ \textsc{lrec}$$

$$\frac{\rho \vdash e \Downarrow v \qquad v.\ell \rightsquigarrow v'}{\rho \vdash e.\ell \Downarrow v'}\ \textsc{rproj} \qquad \frac{\rho \vdash e_1 \Downarrow v_1 \qquad \rho \# v_1 \vdash e_2 \Downarrow v}{\rho \vdash \mathsf{let}^A e_1\ \mathsf{in}\ e_2 \Downarrow v}\ \textsc{letb}$$

$$\frac{\rho \vdash e_1 \Downarrow \{\ell = v'\} \qquad \rho \# v' \vdash e_2 \Downarrow v}{\rho \vdash \mathsf{open}\ e_1\ \mathsf{in}\ e_2 \Downarrow v}\ \textsc{openm} \qquad \frac{\varepsilon \vdash e \Downarrow v}{\rho \vdash \mathsf{struct}^\bullet\, e \Downarrow v}\ \textsc{mstruct-}\bullet \qquad \frac{\rho \vdash e \Downarrow v}{\rho \vdash \mathsf{struct}^\circ\, e \Downarrow v}\ \textsc{mstruct-}\circ$$

$$\frac{}{\rho \vdash \mathsf{functor}^\bullet(A).e \Downarrow \langle \varepsilon, \mathsf{functor}(A).e\rangle}\ \textsc{mfunctor-}\bullet \qquad \frac{}{\rho \vdash \mathsf{functor}^\circ(A).e \Downarrow \langle \rho, \mathsf{functor}(A).e\rangle}\ \textsc{mfunctor-}\circ$$

$$\frac{\rho \vdash e_1 \Downarrow v_1 \qquad \rho \vdash e_2 \Downarrow \langle v_2, \mathsf{functor}(\{\ell : A\}).e\rangle \qquad v_1.\ell \rightsquigarrow v_\ell \qquad v_2 \# \{\ell = v_\ell\} \vdash e \Downarrow v_3}{\rho \vdash e_1 \ltimes e_2 \Downarrow v_1 \# v_3}\ \textsc{mlink}$$

$$\frac{\rho \vdash e_1 \Downarrow v_1 \qquad \rho \vdash e_2 \Downarrow \langle v_2, \mathsf{functor}(D).e\rangle \qquad v_1 \gg_D p \qquad v_2 \# p \vdash e \Downarrow v_3}{\rho \vdash e_1 \ltimes^* e_2 \Downarrow v_1 \# v_3}\ \textsc{mlinkn} \qquad \frac{\rho \vdash e \Downarrow v}{\rho \vdash \mathsf{inl}^B e \Downarrow \mathsf{inl}^B v}\ \textsc{inl}$$

$$\frac{\rho \vdash e \Downarrow v}{\rho \vdash \mathsf{inr}^A e \Downarrow \mathsf{inr}^A v}\ \textsc{inr} \qquad \frac{\rho \vdash e \Downarrow \mathsf{inl}^B v_1 \qquad \rho \# v_1 \vdash e_1 \Downarrow v}{\rho \vdash \mathsf{case}\ e\ \mathsf{of}\ e_1 \mid e_2 \Downarrow v}\ \textsc{case-inl}$$

$$\frac{\rho \vdash e \Downarrow \mathsf{inr}^A v_1 \qquad \rho \# v_1 \vdash e_2 \Downarrow v}{\rho \vdash \mathsf{case}\ e\ \mathsf{of}\ e_1 \mid e_2 \Downarrow v}\ \textsc{case-inr} \qquad \frac{v\ \mathsf{value}}{\rho \vdash \langle v, \mu\lambda(A{\to}B).e\rangle \Downarrow \langle v, \mu\lambda(A{\to}B).e\rangle}\ \textsc{fclos-val}$$

$$\frac{}{\rho \vdash \mu\lambda(A{\to}B).e \Downarrow \langle \rho, \mu\lambda(A{\to}B).e\rangle}\ \textsc{flam} \qquad \frac{\rho \vdash e \Downarrow v}{\rho \vdash \mathsf{fold}^T e \Downarrow \mathsf{fold}^T v}\ \textsc{fold} \qquad \frac{\rho \vdash e \Downarrow \mathsf{fold}^T v}{\rho \vdash \mathsf{unfold}\ e \Downarrow v}\ \textsc{unfold}$$

$$\frac{\rho \vdash e \Downarrow v}{\rho \vdash \langle n\rangle\, e \Downarrow \langle n\rangle\, v}\ \textsc{wrap} \qquad \frac{\rho \vdash e \Downarrow v \qquad v \Downarrow_S^{\mathsf{seal}\, n\, R} w}{\rho \vdash \mathsf{seal}\ n\ R\ S\ e \Downarrow w}\ \textsc{mseal} \qquad \frac{\rho \vdash e \Downarrow v \qquad v \Downarrow_S^{\mathsf{unseal}\, n\, R} w}{\rho \vdash \mathsf{unseal}\ n\ R\ S\ e \Downarrow w}\ \textsc{munseal}$$

2.4 Meta-theory

Definition 2.1 (EVal $\Delta \vdash v \sim_A \overline{v}$).

$$\frac{}{\Delta \vdash i \sim_{\mathsf{Int}} i}\ \text{LIT} \qquad \frac{}{\Delta \vdash \varepsilon \sim_{\varepsilon} \varepsilon}\ \text{UNIT} \qquad \frac{\Delta \vdash v_1 \sim_A \overline{w}_1 \qquad \Delta \vdash v_2 \sim_B \overline{w}_2 \qquad [\![A]\!] * [\![B]\!]}{\Delta \vdash v_1 \# v_2 \sim_{A\&B} \overline{w}_1 \# \overline{w}_2}\ \text{MRG}$$

$$\frac{\Delta \vdash v \sim_A \overline{w}}{\Delta \vdash \{\ell = v\} \sim_{\{\ell:A\}} \{\ell = \overline{w}\}}\ \text{RCD} \qquad \frac{\rho\ \mathsf{value} \qquad \Delta \vdash \rho \sim_{\Gamma'} \overline{\rho} \qquad \Delta; \Gamma' \& A \vdash e : B \rightsquigarrow \overline{e} \qquad [\![\Gamma']\!] * [\![A]\!]}{\Delta \vdash \langle \rho, \lambda A.e \rangle \sim_{A \to B} \langle \overline{\rho}, \lambda([\![A]\!] \to [\![B]\!]).\overline{e} \rangle}\ \text{CLOS}$$

$$\frac{\rho\ \mathsf{value} \qquad \Delta \vdash \rho \sim_{\Gamma'} \overline{\rho} \qquad \Delta; \Gamma' \& A \vdash e : B \rightsquigarrow \overline{e} \qquad [\![\Gamma']\!] * [\![A]\!]}{\Delta \vdash \langle \rho, \mathsf{functor}(A).e \rangle \sim_{\mathsf{sig}\,(A \Rightarrow \langle B \rangle)} \langle \overline{\rho}, \lambda([\![A]\!] \to [\![B]\!]).\overline{e} \rangle}\ \text{MCLOS}$$

$$\frac{\lceil [\![A]\!] \rceil \qquad v\ \mathsf{value} \qquad \Delta; \varepsilon \vdash v : A \rightsquigarrow \overline{w}}{\Delta \vdash v \sim_A [\![A]\!]^{\uparrow}}\ \text{GEN} \qquad \frac{\Delta\, n = R \qquad \Delta \vdash v \sim_R \overline{w}}{\Delta \vdash \langle n \rangle\, v \sim_{\alpha_n} \langle n \rangle\, \overline{w}}\ \text{WRAP}$$

$$\frac{\Delta \vdash v \sim_A \overline{w}}{\Delta \vdash \mathsf{inl}^B v \sim_{A \vee B} \mathsf{inl}^{[\![B]\!]}\, \overline{w}}\ \text{INL} \qquad \frac{\Delta \vdash v \sim_B \overline{w}}{\Delta \vdash \mathsf{inr}^A v \sim_{A \vee B} \mathsf{inr}^{[\![A]\!]}\, \overline{w}}\ \text{INR}$$

$$\frac{\Delta \vdash \rho \sim_{\Gamma'} \overline{\rho} \qquad \Delta; (\Gamma' \& (A \to B)) \& A \vdash e : B \rightsquigarrow \overline{e} \qquad [\![\Gamma']\!] * ([\![A]\!] \to [\![B]\!]) \qquad ([\![\Gamma']\!] \& ([\![A]\!] \to [\![B]\!])) * [\![A]\!] \quad \rho\ \mathsf{value}}{\Delta \vdash \langle \rho, \mu\lambda(A \to B).e \rangle \sim_{A \to B} \langle \overline{\rho}, \mu\lambda([\![A]\!] \to [\![B]\!]).\, \overline{e} \rangle^{[\![B]\!]}}\ \text{FCLOS}$$

$$\frac{\Delta \vdash v \sim_{T[X_0 := \mu T]^{\mu}} \overline{w}}{\Delta \vdash \mathsf{fold}^T v \sim_{\mu T} \mathsf{fold}^{[\![T]\!]}\, \overline{w}}\ \text{FOLD}$$

Theorem 2.2 (seal_type_preservation).

$$\Delta; \Gamma \vdash e : A \rightsquigarrow \overline{e} \implies [\![\Delta]\!]; [\![\Gamma]\!] \vdash \overline{e} : [\![A]\!]$$

Theorem 2.3 (elabSeal_weaken_store).

$$\Delta \sqsubseteq \Delta' \ \wedge\ \Delta; \Gamma \vdash e : A \rightsquigarrow \overline{e} \implies \Delta'; \Gamma \vdash e : A \rightsquigarrow \overline{e}$$

Theorem 2.4 (elabSeal_uniqueness). *Elaboration is a function of the term and (for non-values) its context:*

$$\Delta; \Gamma_1 \vdash e : T_1 \rightsquigarrow \overline{e}_1 \ \wedge\ \Delta; \Gamma_2 \vdash e : T_2 \rightsquigarrow \overline{e}_2 \ \wedge\ (\Gamma_1 = \Gamma_2 \vee e\ \mathsf{value}) \implies T_1 = T_2 \wedge \overline{e}_1 = \overline{e}_2$$

Theorem 2.5 (source_sprogress, source_spreservation, source_bigstep_deterministic). *Pure source metatheory, with elaborability playing the role of typing (SStep is the source small-step, BStep the big-step):*

$$(\exists \overline{e}.\ \Delta; \varepsilon \vdash e : A \rightsquigarrow \overline{e}) \implies e\ \mathsf{value} \vee \exists e'.\ \mathsf{SStep}\ \varepsilon\ e\ e'$$
$$(\exists \overline{e}.\ \Delta; \varepsilon \vdash e : A \rightsquigarrow \overline{e}) \ \wedge\ \mathsf{SStep}\ \varepsilon\ e\ e' \implies \exists \overline{e}'.\ \Delta; \varepsilon \vdash e' : A \rightsquigarrow \overline{e}'$$
$$(\exists \overline{e}.\ \Delta; \varepsilon \vdash e : A \rightsquigarrow \overline{e}) \ \wedge\ \varepsilon \vdash e \Downarrow v_1 \ \wedge\ \varepsilon \vdash e \Downarrow v_2 \implies v_1 = v_2$$

(Each also holds generalized over a well-typed environment; sealed-unit runs are covered — the sealing cases of preservation and progress go through the source coercions.)

Theorem 2.6 (seal_semantic_preservation).

$$\rho \vdash e \Downarrow v \ \wedge\ \Delta; \Gamma \vdash e : A \rightsquigarrow \overline{e} \ \wedge\ \rho\ \mathsf{value} \ \wedge\ \Delta \vdash \rho \sim_{\Gamma} \overline{\rho} \implies \exists \overline{v}.\ \overline{\rho} \vdash \overline{e} \hookrightarrow^* \overline{v} \ \wedge\ \Delta \vdash v \sim_A \overline{v}$$

Theorem 2.7 (seal_whole_program_correctness).

$$\Delta; \varepsilon \vdash e : A \rightsquigarrow \overline{e} \ \wedge\ \varepsilon \vdash e \Downarrow v \implies \exists \overline{v}.\ \varepsilon \vdash \overline{e} \hookrightarrow^* \overline{v} \ \wedge\ \Delta \vdash v \sim_A \overline{v}$$

Theorem 2.8 (seal_separate_compilation).

$$\Delta;\Gamma \vdash e_1 : \Gamma_1 \rightsquigarrow \overline{e}_1 \ \wedge\ \Delta;\Gamma \vdash e_2 : \mathsf{sig}\,(\{\ell : A\} \Rightarrow \langle B\rangle) \rightsquigarrow \overline{e}_2 \ \wedge\ \ell : A \in \Gamma_1 \ \wedge\ [\![\Gamma_1]\!] * [\![\Gamma]\!] \ \wedge\ [\![\Gamma_1]\!] * [\![B]\!] \wedge$$
$$\rho \vdash e_1 \ltimes e_2 \Downarrow v \ \wedge\ \rho\ \mathsf{value} \ \wedge\ \Delta \vdash \rho \sim_\Gamma \overline{\rho}$$
$$\Longrightarrow\ \exists \overline{v}.\ \overline{\rho} \vdash \overline{e}_1 \,\#\, \big((? : [\![\Gamma]\!]) \triangleright (\overline{e}_2\ \{\ell = \overline{e}_1.\ell\})\big) \hookrightarrow^* \overline{v} \ \wedge\ \Delta \vdash v \sim_{\Gamma_1 \& B} \overline{v}$$

Theorem 2.9 (seal_separate_compilation_sealed).

$$\Delta\, n = R \ \wedge\ \mathsf{NoRes}\ [\![R]\!] \ \wedge\ \mathsf{WfSig}\ n\ [\![R]\!]\ [\![\Gamma_1]\!] \ \wedge\ \Delta;\Gamma \vdash e_1 : \Gamma_1[n{:=}R] \rightsquigarrow \overline{e}_1 \ \wedge\ \Delta_c \sqsubseteq \Delta \wedge$$
$$\Delta_c;\Gamma \vdash e_2 : \mathsf{sig}\,(\{\ell : A\} \Rightarrow \langle B\rangle) \rightsquigarrow \overline{e}_2 \ \wedge\ \ell : A \in \Gamma_1 \ \wedge\ [\![\Gamma_1]\!] * [\![\Gamma]\!] \ \wedge\ [\![\Gamma_1]\!] * [\![B]\!] \wedge$$
$$\rho \vdash (\mathsf{seal}\ n\ R\ \Gamma_1\ e_1) \ltimes e_2 \Downarrow v \ \wedge\ \rho\ \mathsf{value} \ \wedge\ \Delta \vdash \rho \sim_\Gamma \overline{\rho}$$
$$\Longrightarrow\ \exists \overline{v}.\ \overline{\rho} \vdash \overline{s} \,\#\, \big((? : [\![\Gamma]\!]) \triangleright (\overline{e}_2\ \{\ell = \overline{s}.\ell\})\big) \hookrightarrow^* \overline{v} \ \wedge\ \Delta \vdash v \sim_{\Gamma_1 \& B} \overline{v} \qquad \textit{where } \overline{s} := \mathsf{seal}\ n\ [\![R]\!]\ [\![\Gamma_1]\!]\ \overline{e}_1$$

SFinitary c is the source-side normalizing fragment (every source former except the fixpoints); elaboration maps it into Finitary (elabSeal_finitary).

Theorem 2.10 (source_representation_independence).

$$\mathsf{OpenBrand}\ [\![\Delta_1]\!]\ [\![\Delta_2]\!]\ \eta\ n\ [\![R_1]\!]\ [\![R_2]\!] \ \wedge\ \Delta_1\, n = R_1 \ \wedge\ \Delta_2\, n = R_2 \wedge$$
$$\mathsf{WfSig}\ n\ [\![R_1]\!]\ [\![S]\!] \ \wedge\ \mathsf{WfSig}\ n\ [\![R_2]\!]\ [\![S]\!] \ \wedge\ \Delta_1;\varepsilon \vdash p_1 : S[n{:=}R_1] \rightsquigarrow \overline{p}_1 \ \wedge\ \Delta_2;\varepsilon \vdash p_2 : S[n{:=}R_2] \rightsquigarrow \overline{p}_2 \wedge$$
$$\mathcal{V}[\![[\![S]\!]]\!]^{(n,[\![R_1]\!],[\![R_2]\!])}(\overline{p}_1, \overline{p}_2) \ \wedge\ \emptyset; S \vdash c : \mathsf{Int} \rightsquigarrow \overline{c} \ \wedge\ \mathsf{SFinitary}\ c \wedge$$
$$\varepsilon \vdash (\mathsf{seal}\ n\ R_1\ S\ p_1) \triangleright c \Downarrow v_1 \ \wedge\ \varepsilon \vdash (\mathsf{seal}\ n\ R_2\ S\ p_2) \triangleright c \Downarrow v_2$$
$$\Longrightarrow\ \exists i.\ v_1 = i \ \wedge\ v_2 = i$$

3 Ongoing work

Planned results are typeset in red; statements are provisional (the source syntax for generics is not yet fixed — a brand-binding $\mathsf{functor}^\beta$ or a separate bfunctor form).

Theorem 3.1 (source_generic_elaboration (planned)). *Source-side generics elaborate to the target $\forall\beta$ layer and are semantically preserved: extending SCE with a brand-binding functor form $\mathsf{bfunctor}(\beta).\, e$ elaborated to $\Lambda_{[\![A]\!]}.\,\overline{e}$ and link-time instantiation elaborated through $\cdot\,[\alpha_n]$, the elaboration type-preservation and simulation theorems (seal_type_preservation, seal_semantic_preservation) extend to the new forms.*

Theorem 3.2 (seal_separate_compilation_generic (planned)). *Separate compilation for generic clients across brand* renaming*: a generic client compiled against an abstract signature links with a provider whose brand was allocated differently — renaming a concrete brand α_n to a fresh $\alpha_{n'}$ in store, derivation and program preserves typing and evaluation (freshness is genuinely required here, unlike for $\forall\beta$-instantiation).*

Theorem 3.3 (source_generic_client_representation_independence (planned)). *The source-level mirror of generic_client_representation_independence: a source generic client, instantiated at each sealed source provider's brand and linked against it, computes the same literal for providers related at the signature — by semantic preservation, the target theorem, and source determinism.*

The concrete work items, in dependency order:

- **Milestone 5 — source-side generics.** Add the brand-binding functor form to SCE, its elabSeal rules targeting $\Lambda/\cdot\,[\alpha_n]$, and the Correctness cases; instantiation happens at link time ($\ltimes^*$).
- **Brand-renaming metatheory.** Renaming concrete brand → concrete brand with freshness side conditions, through stores, derivations and values; culminates in seal_separate_compilation_generic. A separate mechanism from $\forall\beta$-instantiation, which is freshness-free by design.
- **Source-level generic example.** The $\{mk\}\,\&\,\{get\}$ example with a source ∀-functor client serving both sealed source providers — the source mirror of the mechanized target-level run.
- **μ/fix examples.** A sealed-interface factorial through the fixpoint closure, a tagged-union run, and a μ-stream, exercising the Phase-1 layers end-to-end in Examples.lean.

- **Step-indexed logical relation.** Lift the Finitary/SFinitary fence so the relational results (and RI) cover fixpoints and a structural μ-clause; replaces the termination-flavored fundamental lemma with a step-indexed one.
- **Lifting the $\forall$-fences.** A shifting μ-substitution admits $\forall$ inside μ-bodies ($\neg\forall\in T$ on fold/unfold); shifting in the term-level instantiation admits nested, multi-binder generics (BClosed 1 A on TBAPP).
- **Relational parametricity proper.** Free theorems over arbitrary admissible relations at β need brand-interpretation environments plus a properly-bound (EBClosed) typing discipline; representation independence of generic clients deliberately avoided both.
- **Cleanups.** Collapse the module-type syntax into the function type, merge single-import linking into the n-ary form, and update this document as the source syntax for generics lands.